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PLATING POSTERS INC · METAL FINISHING REFERENCE SERIES

ACID COPPER

THE COMPLETE TRAINING MANUAL

A full training course for the brand-new operator — from "what is an anode?" to a signed certificate of completion. Acid copper sulfate plating: the best-leveling, highest-throwing bath in the shop, and the leveling heart of decorative nickel-chrome and PCB copper. Assumes you know nothing. Teaches you everything.

01 CLEAN ▸	02 RINSE ▸	03 ACTIVATE ▸	04 STRIKE
05 PLATE ▸	06 RINSE ▸	07 POST-TREAT ▸	08 RINSE/DRY

OPERATOR TRAINING & CERTIFICATION

EDITION 1.0 · 2026 · BEGINNER-FRIENDLY STYLE

Training reference only. Always verify exact parameters against your chemistry supplier's Technical Data Sheets (TDS), customer specifications, current Safety Data Sheets (SDS), and applicable OSHA, EPA, REACH, and local regulations before production use.

FIND YOUR WAY

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★ **REMEMBER**

This manual is a training and reference tool — **not** a substitute for your facility’s written procedures, your supplier’s TDS/SDS, or your supervisor’s instructions. When this manual and your shop’s official procedure disagree, **follow your shop’s official procedure and ask your supervisor.**

PART ONE — FRONT MATTER & FUNDAMENTALS

START HERE

WELCOME ABOARD

IF YOU DON'T KNOW AN ANODE FROM AN APRON, YOU ARE IN EXACTLY THE RIGHT PLACE

Welcome. If you are holding this manual, you are about to learn how to run — or at least understand — an **acid copper plating line**. Maybe you started this week. Maybe you have been racking parts for a year and finally want to know *why* you do the things you do. Either way, you are in the right place, and we are glad you are here.

Here is the most important thing we can tell you up front: **this manual assumes you know nothing about plating, chemistry, or electricity.** That is not an insult. That is a promise. Every term gets defined the first time we use it. Every step explains not just *what* to do but *why* it matters. Nobody is born knowing what "throwing power" means or why a trace of chloride you can barely measure can make or break a whole tank. You learn it. We are going to teach it.

This manual is written in the spirit of the *plain-language* guides — friendly, plain-spoken, and patient. We would rather over-explain than leave you guessing in front of a tank full of acid.

> ★ **REMEMBER**
> You do not need to memorize this manual. You need to *understand* it. Understanding sticks. Memorizing fades by Friday.

• HOW TO USE THIS MANUAL

The manual follows the **line itself**, station by station, in the order a part actually travels: clean it, rinse it, activate it, **strike it**, plate it, rinse it, post-treat/inspect it, and rinse/dry it. Reading it in order matters, because (as you'll soon learn) **the order of the plating line is not random** — and neither is the order of this book. Acid copper adds one stage most beginners haven't met: a **copper strike**, the thin flash that lets copper stick to steel at all.

THE ICONS — YOUR FRIENDLY MARGIN HELPERS

As you read, you will run into little flagged notes. Here is what each one means.

> 💡 **TIP**
> A shortcut, a trick of the trade, or a smarter way to do something — the kind of thing a 20-year veteran would lean over and tell you. These save you time and headaches.

> ★ **REMEMBER**
> A core idea worth locking in. If you forget everything else on the page, don't forget these. They're the load-bearing ideas.

> ⚠ **HEADS-UP**
> A safety warning. Read every single one. These are the notes that keep your skin, your eyes, and your lungs intact. We use them *liberally* because an acid copper line — especially one with a cyanide strike upstream — has real, in some cases lethal, hazards.

> ⚙ **TECHNICAL STUFF**
> Deeper chemistry or theory for the curious. Totally optional. Skip these and you can still run the line perfectly. Read them and you will understand it on a deeper level.

> ☀ **FUN FACT**
> A bit of interesting trivia to make the science stick (and to give you something to say at lunch).

TWO MORE THINGS YOU'LL SEE AT EVERY STATION

Besides the margin icons, each station chapter ends with two recurring tools. The first is a **Teach-Back** checklist (sometimes paired with a red "Top Mistakes" card) — the things you must be able to *say out loud, unprompted* before you're cleared on that station; it's how you and your trainer confirm the ideas actually stuck. The second is a bordered **Sign-Off** box at the very end of the station — the short statement your *trainer initials* (with you) once you've demonstrated the skill safely. Treat the Teach-Back as your study guide and the Sign-Off as the gate you have to pass through.

WHAT YOU'LL BE ABLE TO DO

LEARNING OBJECTIVES

THE WHOLE FOUNDATION — MASTER THIS AND EVERY STAGE CHAPTER CLICKS INTO PLACE

By the time you finish this manual, you should be able to confidently:

1. **Explain what electroplating is** in plain language — how electricity moves metal from one place onto another.
2. **Describe what acid copper plating does** and why a shop chooses it — best-in-class leveling, high throwing power, near-100% efficiency, and a soft, ductile deposit.
3. **Explain why copper cannot go straight onto steel, zinc, or aluminum** — and what a *strike* is for.
4. **Name all eight stages of the line in order** and explain *why* the order cannot be rearranged.
5. **Understand what each tank is for** — what goes in, what comes out, and what "good" looks like at every handoff.
6. **Recognize the main hazards** at each station — acids, alkali, electrical, acid mist, copper-sulfate toxicity, and (where a cyanide strike is used) the lethal acid-plus-cyanide hazard — and know the correct PPE and emergency response for each.
7. **Use the basic field tests** every operator relies on, like the water-break test, chloride titration, and the Hull Cell.
8. **Spot trouble early** — read a deposit and know whether the problem is upstream, in the bath, in the anodes, or in your handling.
9. **Talk the language** — use the right words so you can communicate clearly with your lead, your chemist, and your chemistry supplier.

> ★ **REMEMBER**
> You are not expected to hit all of these on day one. Plating is a craft. The goal is steady competence, not instant mastery.

THE EIGHT-STAGE PROCESS AT A GLANCE

Here is the whole acid copper line in one strip. Don't memorize it yet — just get a feel for the rhythm: **Clean → Rinse → Activate → Strike → Plate → Rinse → Post-Treat/QC → Rinse/Dry.**

#	STAGE	THE ONE THING IT DOES	TEMP	TIME
01	Soak Clean / Electroclean	Removes all oil, grease, and shop soil	130–180°F	3–10 min
02	Rinse (Pre-Activation)	Washes off cleaner before the acid dip	Ambient	30–60 s
03	Acid Activation (Acid Dip)	Dissolves surface oxide; exposes bare metal	Amb–110°F	5–60 s
04	Copper Strike	Thin flash that stops immersion copper on active metals	90–140°F	1–5 min
05	Acid Copper Plating	Deposits the bright, leveled copper	70–90°F	5–60 min
06	Rinse (Post-Plate)	Recovers copper drag-out; protects downstream	Ambient	30–60 s
07	Post-Treatment / QC	Hand off to Ni-Cr, protect bare copper, or verify	Varies	Varies
08	Final Rinse / Dry	Cleans up and dries without spotting the copper	Amb–140°F	30–60 s + dry

> ★ **REMEMBER**
> Notice the pattern: **a rinse follows almost every working tank.** Rinses are not breaks — they are checkpoints that keep one chemistry from poisoning the next. On an acid copper line one of those handoffs is special: the rinse between the *acid* activation and a *cyanide* strike is not just a quality checkpoint — it is a **life-safety firewall**, because acid meeting cyanide releases lethal gas.

CHAPTER 1

WHAT IS ELECTROPLATING?

THE TRUE BEGINNER'S PRIMER — SO FAR BACK WE DON'T EVEN ASSUME YOU KNOW WHAT PLATING IS

THE ONE-SENTENCE VERSION

Electroplating is using electricity to coat one metal with a thin layer of another metal. That's the whole idea. Now let's unpack it so it actually means something.

Imagine you have a steel part with tiny polishing scratches you want to hide, and you want it to end up smooth, bright, and ready for nickel and chrome. You want to give it a thin layer of **copper** — a soft, ductile metal that flows into scratches and levels the surface better than almost anything else. You *could* try to polish forever, but copper does the leveling for you. We use electricity to grow a perfectly even, tightly-bonded layer of copper directly onto the part, atom by atom. That process is electroplating.

HOW ELECTRICITY MOVES METAL: THE KITCHEN-SINK PICTURE

Here is how it physically works. Picture a tank full of liquid called the **electrolyte** (or just "the bath"). This liquid has copper dissolved in it — invisible copper, broken down into tiny charged particles called **ions**. An *ion* is just an atom that carries an electrical charge. Dissolved copper floats around the bath as positively-charged copper ions, written **Cu^{2+}**.

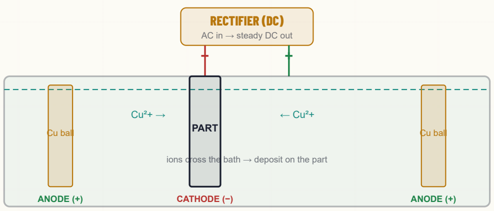
You hook two things up to a power supply called a **rectifier** (it converts ordinary AC wall power into the steady one-direction "DC" current plating needs):

- **The cathode** — this is your *part*, connected to the **negative** terminal. Just remember "cathode = the part, the thing collecting metal."
- **The anode** — bars or balls of copper hanging in the same tank, connected to the **positive** terminal. As the bath uses up its dissolved copper, the anodes slowly dissolve to replace it — they are the "fuel tank" of copper metal. ("Anode = positive = the metal that feeds the bath.")

When you turn on the rectifier, here is the dance that happens:

1. Current flows. The positively-charged copper ions (Cu^{2+}) in the bath are *attracted* to the negatively-charged part (the cathode), because opposite charges attract.
2. When a copper ion reaches the part's surface, it grabs electrons from the part and turns back into solid copper metal, sticking to the surface.
3. Meanwhile, at the anodes, solid copper dissolves *into* the bath, releasing new Cu^{2+} ions to keep the supply topped up.

The net result: copper moves off the anodes, through the bath, and onto your part. The longer you plate and the more current you push, the thicker the coating.



★ REMEMBER

Three words to lock in now: **Cathode = the part** (collects copper). **Anode = the copper balls** (feed the bath). **Electrolyte = the bath** (carries copper from one to the other). Almost everything in plating is a variation on this one idea.

⚙ TECHNICAL STUFF

At the part, a **reduction**: $Cu^{2+} + 2e^- \rightarrow Cu^0$. At the anode, the reverse — solid copper gives up two electrons and dissolves: $Cu^0 \rightarrow Cu^{2+} + 2e^-$. Because the anodes replenish what the cathode consumes, a well-run acid copper bath is largely self-balancing on copper metal — you mostly maintain acid, chloride, and additives, not the copper itself.

☀ FUN FACT

The unit "volt" traces back to Alessandro Volta, whose first battery (the "voltaic pile," 1800) was a stack of copper and zinc discs. Copper has been at the heart of electricity since literally the first battery — and you'll read volts on your rectifier every day.

⚠ HEADS-UP

Plating rectifiers push enormous current — hundreds to thousands of amps — low voltage but very high current, into a tank full of conductive liquid. **Never reach into an energized tank, and never do maintenance until equipment is locked and tagged out (LOTO).** Electricity plus acid plus your hand demands total respect.

CHAPTER 2

WHAT IS "ACID COPPER"?

AND WHY DO WE USE IT — THE BEST-LEVELING, HIGHEST-THROWING BATH IN THE SHOP

COPPER PLATING COMES IN FAMILIES

"Copper plating" just means coating something with copper. But there are *several different recipes* for the bath that does it. The three families you'll hear about are:

- **Cyanide copper** — an alkaline (high-pH) bath that dissolves copper using cyanide. It has excellent adhesion and coverage on bare steel, which is why it's still used as a *strike* — but cyanide is a **lethal poison** and demands strict segregation from acids.
- **Alkaline non-cyanide copper** — a modern high-pH bath (often pyrophosphate- or complex-based) that avoids cyanide entirely. Used as a cyanide-free strike and buildup bath.
- **Acid copper (sulfate)** — a low-pH bath of copper sulfate and sulfuric acid. This manual is about acid copper. It gives the best leveling and throwing power of the common baths — but it will *not* plate directly on active metals like steel, so it needs a strike first.

⚙ **TECHNICAL STUFF**

pH is the scale we use to measure how acidic or basic a liquid is. It runs from 0 (strong acid) to 14 (strong base), with 7 being neutral (pure water). Lower number = more acidic. Our acid copper bath is **strongly acidic** from its sulfuric acid content — nothing like the mild acidity of a coffee; treat it as a real mineral-acid bath. The cyanide and alkaline-non-cyanide *strikes*, by contrast, are strongly *basic*.

More precisely, our process is **acid copper sulfate**. The bath is built from just three simple ingredients — **copper sulfate ($CuSO_4 \cdot 5H_2O$)** for the metal, **sulfuric acid (H_2SO_4)** for conductivity and throwing power, and a trace of **chloride (Cl^-)** — plus an organic additive package. The metal chemistry is inexpensive and well understood. The *control* is where the skill lives.

THE BATH AT A GLANCE

COMPONENT	TYPICAL RANGE	WHAT IT DOES
Copper sulfate ($CuSO_4 \cdot 5H_2O$)	180–250 g/L (24–33 oz/gal)	The metal you're plating (decorative range)
Sulfuric acid (H_2SO_4)	45–75 g/L (6–10 oz/gal)	Conductivity, throwing power, anode corrosion
Chloride ion (Cl^-)	50–80 ppm	Anode film control; works with suppressor to refine the deposit
Carrier / suppressor	Per supplier's TDS	Polarizes the surface, refines grain, adds luster (needs chloride)
Brightener / accelerator	Per supplier's TDS	Brightness, throw, via-fill (e.g. SPS)
Leveler	Per supplier's TDS	Smooths the surface; controls low-current areas
Temperature	70–90°F (target ~75–80°F)	Ambient bath; often needs cooling, not heating

⚙ **TECHNICAL STUFF**

"g/L" means **grams per liter** — how many grams of a substance are dissolved in each liter of bath. "oz/gal" is the same idea in ounces per gallon, which many US shops still use. "ppm" is **parts per million** — used for the tiny chloride level. A "**TDS**" is a **Technical Data Sheet** — the instruction document your chemistry supplier provides for each product. When this manual says "per TDS," it means *follow the supplier's sheet*, because additive amounts are specific to each product line. Never guess at additive levels.

★ **REMEMBER**

The metal chemistry of acid copper is simple — copper sulfate, sulfuric acid, a pinch of chloride. What makes it a craft is holding the **chloride, the additives, and the anodes** in balance. Those three are the whole game, and we'll come back to them again and again.

• WHY DO SHOPS USE ACID COPPER?

Four big reasons, and you should be able to name them: **leveling**, **throwing power**, **high efficiency with a soft, ductile deposit**, and a **cheap, non-cyanide main bath**. Let's take them one at a time.

REASON 1: BEST-IN-CLASS LEVELING

Leveling means the deposit fills in tiny scratches, tooling marks, and polishing lines — it comes out *smoother than the surface it went onto*. Acid copper, with a balanced additive system, levels better than any other common plating bath, including nickel. That is the whole reason acid copper is the preferred **decorative undercoat**: it does the smoothing work so the nickel and chrome on top look flawless over an imperfect substrate.

REASON 2: EXCELLENT THROWING POWER

"Throwing power" is the bath's ability to plate evenly into deep recesses, blind holes, through-holes, and the insides of tubes. High-acid / low-copper formulations of acid copper throw *extremely* well — which is exactly why acid copper is the backbone of **printed circuit board (PCB)** copper, plating uniformly down through-holes and filling blind vias.

REASON 3: HIGH EFFICIENCY, SOFT & DUCTILE DEPOSIT

Acid copper runs at roughly **95–100% cathode efficiency** — nearly every amp you pay for deposits copper, with very little wasted making hydrogen gas. And a properly additive-balanced deposit is **soft, low-stress, and ductile** — ideal for parts that will be formed afterward, and for the thermal-cycling reliability that PCBs demand.

> ⚙ **TECHNICAL STUFF**
>
> **Cathode efficiency** is the percentage of your current that actually deposits metal, versus being wasted (mostly splitting water into hydrogen gas). Acid copper's ~95–100% is about as good as plating gets. It also means acid copper plates fast — high-copper formulations build thick deposits quickly for buildup and electroforming.

REASON 4: CHEAP, SIMPLE, NON-CYANIDE MAIN BATH

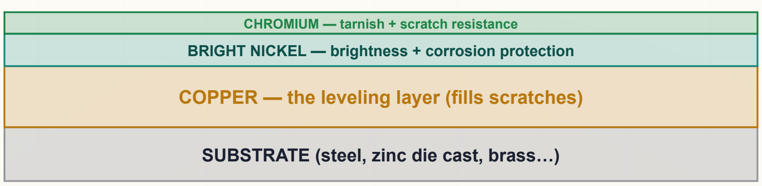
Copper sulfate + sulfuric acid + trace chloride is inexpensive and well characterized, and — unlike cyanide copper — the *main* acid copper bath contains no cyanide, so its waste treatment is far simpler. (Note the word "main": if a *cyanide strike* is used upstream, cyanide handling still applies to that tank.)

• WHERE ACID COPPER FITS: THE DECORATIVE TRIAD

You'll constantly hear acid copper described as part of a **triad**. The classic decorative finish on auto trim, plumbing fixtures, and hardware is built in three plated layers, per **ASTM B456**:

THE DECORATIVE TRIAD (NOT TO SCALE)

COPPER LEVELS · NICKEL PROTECTS · CHROME FINISHES



> ★ **REMEMBER**
>
> **Copper (leveling) → Bright Nickel (brightness + corrosion) → Chromium (tarnish/scratch resistance).** Copper's leveling is what lets the whole system look mirror-perfect over a rough substrate. That's why the decorative world leans on acid copper as the *foundation* of the finish.

• THE RULE EVERYONE LEARNS THE HARD WAY: STRIKE FIRST

Here is the single most important fact about acid copper, and it trips up every beginner: **you cannot plate acid copper directly onto steel, zinc die cast, or aluminum.** Copper is a *noble* (unreactive) metal, and steel, zinc, and aluminum are *active* (reactive) metals. Put an active metal into an acid copper bath and copper deposits on it **instantly — before you even turn on the current —** by a purely chemical reaction called **immersion** (or galvanic displacement) deposition.

> ★ **REMEMBER — THE IMMERSION REACTION**
>
> **$Cu^{2+} + Fe^0 \rightarrow Cu^0 + Fe^{2+}$.** The copper ions in the bath rip electrons off the steel and slap themselves onto the surface as loose, powdery, *non-adherent* copper — while iron dissolves into the bath. That copper does not stick. Plate over it and you get **blistering, not brightness.** This is why the strike exists.

The fix is the **copper strike** (Station 04): a thin, tightly-bonded flash of copper (or nickel) laid down under controlled conditions — from a cyanide, alkaline non-cyanide, or Woods nickel bath — so that by the time the part reaches the acid copper tank, it presents a *compatible, noble copper surface* the acid copper can build on properly. On copper and brass substrates (which are already noble enough) a strike can sometimes be skipped after excellent activation, but it still improves adhesion. **On steel, zinc die cast, and aluminum the strike is mandatory.** We devote all of Station 04 to it.

> 💡 **TIP**
>
> If anyone ever tells you “just dip the steel part in the copper tank,” stop. That’s how you grow blisters. Active metals get a strike first — every time.

• THE CHLORIDE WINDOW: 50–80 PPM

Chloride is present at only **parts-per-million** — a level you can barely measure — yet it is the single most sensitive routine control in the whole bath. It partners with the suppressor to refine the deposit, and it keeps the copper anodes dissolving smoothly. Hold it in the window:

CHLORIDE LEVEL	WHAT HAPPENS
Below ~40 ppm (too low)	Anodes polarize; deposit goes dull, coarse, and columnar; poor throw
50–80 ppm (decorative target)	Bright, leveled, well-behaved deposit; healthy anode film
Above ~100 ppm (too high)	Anodes passivate (green/gray film), shed copper powder → roughness; dull edges

PCB / high-throw baths run a slightly lower window (**40–60 ppm**). Either way, you **titrate chloride with silver nitrate** — you measure it, you don’t guess it.

• THE ADDITIVE TRIAD: SUPPRESSOR, ACCELERATOR, LEVELER

The organic additive package is a **three-part team**, and each member does a different job. You’ll meet them in depth at Station 05; here’s the overview:

ADDITIVE	PLAIN-LANGUAGE JOB	NOTE
Carrier / Suppressor	Polarizes the surface, refines grain, gives luster and macro-throw	**Only works with chloride present** — the two are a pair
Brightener / Accelerator	Speeds deposition in high-current areas — brightness, throw, via-fill	e.g. SPS; builds up unless consumed/dragged out
Leveler	Smooths the surface, controls low-current areas, prevents “dog-boning”	Consumed at the surface; nitrogen-bearing

> ★ **REMEMBER**
>
> **Suppressor and chloride work as a pair — neither performs well without the other.** That’s why a “dull, coarse” deposit sends you to check *both* your chloride titration and your suppressor level.

• THE ANODES HAVE A SECRET: THE PHOSPHORUS FILM

Acid copper uses **phosphorized copper anodes** — copper alloyed with a little phosphorus (**0.02–0.08% P**). As they work, they grow a thin **black film** on their surface (it contains a compound called Cu_3P). That film is not dirt — it is essential. It makes the anodes dissolve smoothly and evenly and suppresses the formation of copper powder.

ANODE STATE	CAUSE	RESULT ON YOUR PARTS
Healthy black film	Right phosphorus, chloride in range	Smooth, uniform copper dissolution — good
Filmless anode	Freshly loaded / not conditioned; chloride too low	Sheds copper particles → **roughness**
Passivated anode	Chloride too high, current too high, too little anode area	Stops dissolving → copper depletes, voltage climbs → **starvation**

Anodes are loaded as balls or chunks into **titanium baskets** and enclosed in **bags** so any fines they shed can't reach your parts. Condition new anodes and hold chloride in range to keep that film healthy.

• TWO REGIMES: THE COPPER-TO-ACID LEVER

The *same* family of bath is tuned two very different ways depending on the work, and the master lever is the **ratio of copper to acid**:

REGIME	COPPER	ACID	BEST AT
Decorative (high copper / low acid)	$CuSO_4 \cdot 5H_2O$ 180–250 g/L	H_2SO_4 45–75 g/L	Fast plating, brightness, leveling
PCB / high-throw (low copper / high acid)	Cu 60–120 g/L	H_2SO_4 180–240 g/L	Superior throwing power, uniform distribution

★ **REMEMBER**

The **copper : acid ratio is the master lever.** High copper / low acid = faster plating and good luster (decorative). Low copper / high acid = superior throwing power (PCB, complex geometry). Never move one without thinking about the other — move them together toward the target regime for the work in front of you.

• THE HONEST TRADEOFFS

A good operator knows the weaknesses too. Acid copper is superb, but it is *demanding*:

- **Will not plate directly on active metals.** Steel, zinc, and aluminum immersion-deposit copper and blister — a strike is required. (This is the big one.)
- **Extremely sensitive to particulate.** Dirt is the #1 cause of roughness and nodules — the bath demands aggressive, continuous filtration.
- **The chloride window is narrow (50–80 ppm).** Too low or too high both wreck the deposit and the anodes.
- **The organic additives break down** and must be replenished and periodically *carbon-treated*.
- **Needs phosphorized anodes with a maintained film** — filmless makes powder, passivated makes starvation.
- **Copper sulfate is highly toxic to aquatic life** — strict wastewater limits.
- **Sulfuric acid mist** from air agitation requires ventilation and mist control. And if a **cyanide strike** is used upstream, its hazards apply too.

A WORD ABOUT WHAT COMES AFTER: POST-TREATMENT

Unlike zinc, copper rarely stands alone. Bright bare copper **tarnishes within a day** in air. So the line usually ends by handing the copper off to **bright nickel + chrome** (the decorative triad), or by protecting a bare-copper finish with an **anti-tarnish** or lacquer, or — for PCBs — by moving to the next fabrication step. We cover all of these at Station 07.

★ **REMEMBER**

Simple metal chemistry; ruthless on chloride, particulate, and anode film. Control those three and the copper is flawless. That one sentence is most of what makes an acid copper operator good at the job.

CHAPTER 3

HOW A PLATING LINE WORKS

THE BIG PICTURE — A LINE IS A SYSTEM, NOT A PILE OF SEPARATE TANKS

A LINE IS A SEQUENCE OF TANKS

An acid copper line is not one tank. It's a **sequence of tanks** that a part travels through in a strict order. Think of it like a kitchen line or a car wash: each station does one job, and each job prepares the surface for the next. Parts ride down the line in one of two ways:

- **Rack plating** — Parts are individually hung on a **rack** (a frame with hooks or clips that holds parts and carries electrical current to them). Used for larger, cosmetic, or delicate parts — and for most decorative copper.
- **Barrel plating** — Lots of small parts are dumped together into a perforated rotating **barrel** that tumbles them through the chemistry. Far more efficient for high volumes of tiny parts; runs at lower current density.

TIP
You'll see different numbers for the same chemistry depending on rack vs. barrel — and a whole different regime again for PCB / high-throw work. When in doubt about which numbers apply, check whether you're running rack, barrel, or board, and read the matching column on your shop-floor poster.

THE WHOLE ACID COPPER SEQUENCE: 8 STAGES

Here is the complete process. Don't memorize the details — just get a feel for the *flow.* Each stage gets its own full chapter later.

#	STAGE	THE ONE THING IT DOES	TEMP	TIME
01	Soak Clean / Electroclean	Removes all oil, grease, and shop soil	130–180°F	3–10 min
02	Rinse (Pre-Activation)	Washes off cleaner before the acid dip	Ambient	30–60 s
03	Acid Activation (Acid Dip)	Dissolves surface oxide; "wakes up" the metal	Amb–110°F	5–60 s
04	Copper Strike	Thin flash that stops immersion copper on active metals	90–140°F	1–5 min
05	Acid Copper Plating	Deposits the bright, leveled copper	70–90°F	5–60 min
06	Rinse (Post-Plate)	Recovers copper drag-out; protects downstream	Ambient	30–60 s
07	Post-Treatment / QC	Hand off to Ni-Cr, protect bare copper, or verify	Varies	Varies
08	Final Rinse / Dry	Cleans up and dries without spotting the copper	Amb–140°F	30–60 s + dry

The basic rhythm: **Clean** → **Rinse** → **Activate** → **Strike** → **Plate** → **Rinse** → **Post-Treat** → **Rinse/Dry.**

A LINE IS A SYSTEM, NOT A PILE OF SEPARATE TANKS

★ **REMEMBER**
Every stage either protects the next stage or poisons it. There is no neutral step. A part that leaves the cleaner still carrying a film of cleaner will contaminate the acid. A part that leaves the acid still wet with acid will foul the strike. What you do (or fail to do) at Stage 1 shows up as a defect at Stage 5. The line is a chain, and a chain is only as strong as its weakest link.

TIP
Think of the acid copper tank as the crown jewel of the line. Everything *before* it (clean, rinse, activate, strike) exists to hand it a perfectly clean, struck, well-rinsed part. Everything *after* it (rinse, post-treat, rinse/dry) exists to protect and finish what the copper created. The whole line orbits that one tank — and its three obsessions: particulate, chloride, and anode film.

CHAPTER 4

WHY THE ORDER MATTERS

WALKING THE LINE — EACH STAGE SETS UP THE NEXT, AND THE SEQUENCE IS LOCKED BY CHEMISTRY

You might think the stage order is just convention — that you could shuffle steps or skip one to save time. You can't. Each stage exists *because of* the stage before and after it. Here is the logic, link by link.

STAGE 1 — CLEAN: GET RID OF THE JUNK FIRST

Plating, acid, and the strike all need to touch *bare metal*. Oil and shop soil block them. **Plate over dirt and you've just plated dirt** — and because copper *levels and brightens*, it magnifies every cleaning miss as a pit, a skip, or an adhesion failure. There is a free, foolproof test for cleanliness called the **water-break test** (see the cleaning chapter): if water sheets off in an unbroken film, the part is clean; if it beads up like on a waxed car, oil remains.

> 💡 **TIP**
> The simplest, most powerful quality check in the whole shop is the **water-break test**, and it's free. Pull a cleaned part and watch how water sheets off it. A smooth, continuous, unbroken sheet ("zero seconds" to break) means the surface is clean. If water beads up or "breaks" into droplets, there's still oil there — *stop and re-clean.*

> ⚠ **HEADS-UP**
> The cleaner is **hot (130–180°F) and strongly alkaline (pH 11–13.5)** — it causes chemical burns and can seriously injure your eyes. Electrocleaning also generates hydrogen and oxygen gas — ventilation required, no open flames. Wear full PPE, and know where your shower and eyewash are before you need them.

STAGE 2 — RINSE (PRE-ACTIVATION): DON'T CARRY THE CLEANER FORWARD

The cleaner is alkaline (basic). The next tank is acid. If you carry cleaner straight into the acid, it neutralizes the acid, wastes it, and weakens the activation. The rinse strips the cleaner film off first.

> ★ **REMEMBER**
> *"Drag cleaner into acid and you waste acid. Drag cleaner into the strike and you kill adhesion."* Rinses exist to keep each chemistry from invading the next tank. The cleaner-to-acid rinse is your first firewall on the line.

STAGE 3 — ACID ACTIVATION: WAKE UP THE METAL

Even a clean metal surface grows an invisible layer of **oxide** (a tarnish-like film from reacting with air). Copper (or the strike) can't bond to oxide. The acid dip dissolves that oxide and leaves a chemically "fresh," active surface for the strike to grip.

- **Under-dip** (too short/weak) = oxide remains = poor adhesion later.
- **Over-dip** (too long/strong) = you etch and pit the part (and on brass, *dezincify* it), and on high-strength steel you load it with hydrogen.

> ⚠ **HEADS-UP — A LIFE-SAFETY FIREWALL**
> On an acid copper line the rinse *after* the acid activation is doubly critical, because the next tank may be a **cyanide strike**. **Acid dragged into a cyanide bath — or acid-wet parts entering it — can release hydrogen cyanide (HCN) gas, which can be fatal.** The rinse and drain between acid and a cyanide strike is not just a quality step; it is a *life-safety* step. Never let acid and cyanide meet.

> ⚙ **TECHNICAL STUFF**
> **Hydrogen embrittlement** is a real, serious risk on hard steel. During pickling and plating, atomic hydrogen can diffuse into steel's crystal structure. In high-strength steel (above ~40 HRC, or 180 ksi tensile), trapped hydrogen can cause catastrophic cracking later, under load. The fix is **baking**: 375 ±25°F for at least 8 hours, started **within 4 hours** of the last electrodeposition step (per ASTM B850 `[VERIFY rev]`). For prone parts, use *inhibited* acids and minimum dip time.

STAGE 4 — COPPER STRIKE: THE BRIDGE ONTO ACTIVE METAL

This is the stage that has no equivalent on a simple zinc line, and it's the reason acid copper works at all on steel. Copper is more noble than steel, zinc, and aluminum, so those metals would **immersion-plate** loose, blistery copper the instant they hit the acid copper tank. The strike lays down a thin, tightly-bonded copper (or nickel) flash first — under alkaline (cyanide or non-cyanide) or Woods-nickel conditions where immersion doesn't happen — giving the acid copper a compatible surface to build on.

> ★ **REMEMBER**
> **No strike, no adhesion on active metals. Period.** Enter the strike *live* (current already on), get **complete coverage**, then rinse before acid copper. A missed or incomplete strike is the number-one cause of blistering downstream.

STAGE 5 — ACID COPPER PLATING: THE MAIN EVENT

Now — and only now, with a clean, activated, struck, well-rinsed part — does the real copper go on. The bath is acidic sulfate chemistry running near room temperature (70–90°F), with phosphorized copper anodes feeding fresh metal back into solution as you plate. This is where leveling and brightness happen.

> ⚠ **HEADS-UP**
> The plating bath is **sulfuric-acid-strong** and air agitation throws up an **acid / copper-sulfate mist** you must not breathe — local exhaust ventilation is required. The rectifier runs **high amperage** — shock and arc-flash risk. Copper sulfate is highly toxic to aquatic life. LOTO before any rectifier work.

STAGE 6 — RINSE (POST-PLATE): RECOVER THE COPPER, PROTECT DOWNSTREAM

The freshly plated part drips acid copper — which is both **money** (copper you paid for) and a **contaminant** to whatever comes next. A drag-out recovery rinse captures copper for return to the bath; the clean rinse then keeps copper and acid out of the downstream nickel or post-treatment tank.

> ★ **REMEMBER**
> **Copper dragged into the nickel tank is a dollar lost *and* a defect waiting to happen.** Even a few ppm of copper makes dark, pitted low-current areas in a nickel bath. This rinse does double duty: cost control and quality control.

STAGE 7 — POST-TREATMENT / QC: DECIDE WHAT COPPER DOES NEXT

Copper rarely stands alone. This stage is where the part is either **handed off to bright nickel + chrome** (the decorative triad), **protected** as a bare-copper finish (anti-tarnish, antique blackening, or lacquer), or **verified and finished** as a functional/PCB deposit. It's also the decision point for the **hydrogen-embrittlement relief bake** on hardened steel.

STAGE 8 — FINAL RINSE / DRY: SEND IT OUT RIGHT

A last clean DI rinse removes residue, and a *gentle* warm-air dry finishes the part without water spots. Fresh copper oxidizes fast — keep it protected, handle with clean gloves, and don't let it sit wet.

> ★ **REMEMBER**
> The whole eight-stage sequence is one continuous logic chain: *clean it so copper can stick → rinse so the cleaner doesn't ruin the acid → activate so the metal is bare and ready → strike so acid copper has a noble surface to build on → plate the bright, leveled copper → rinse to recover copper and protect downstream → post-treat/verify → final rinse and gentle dry so it ships flawless.* Every step earns its place. Skip one, and the chemistry of the next step tells on you.

> 💡 **TIP**
> When you understand *why* each stage exists, troubleshooting becomes logical instead of mysterious. Blistering? Look upstream at the strike, cleaning, and activation. Copper climbing in the nickel bath? Look at the post-plate rinse. The defect almost always points back to the stage that was supposed to prepare for it.

CHAPTER 5

PPE — YOUR DAILY ARMOR

THE GEAR THAT STANDS BETWEEN YOU AND CHEMICALS THAT BURN, MIST THAT HARMS, AND — ON THE STRIKE — GAS THAT KILLS

Now the part that keeps you safe. An acid copper line works with hot caustic, strong mineral acids, electrical current at high amperage, acid and copper-sulfate mist, and — if a cyanide strike is used — one of the deadliest hazards in all of metal finishing. None of these will hurt you if you respect them and wear the right gear. All of them can hurt you badly if you don't.

★ **REMEMBER**
There is no part, no order, and no deadline worth an injury. Scrap can be reworked. Your eyes — and your lungs — cannot.

THE MASTER PPE SET (WEAR AT ALL STATIONS)

Treat this as your baseline "uniform" any time you are on the line:

- **Chemical splash goggles** — sealed goggles, not safety glasses. Glasses leave gaps; a splash finds the gaps.
- **Face shield** — *over* the goggles for any work involving acids, hot caustic, the strike, or the plating bath. The shield protects your face; the goggles are your eye backup.
- **Chemical-resistant gloves** — elbow-length, rated for the chemistry (nitrile, neoprene, or PVC). Inspect for pinholes before each use.
- **Chemical-resistant apron** — full-length, covering chest to below the knee.
- **Chemical-resistant boots** — closed, non-slip, splash-proof. Floors near plating tanks get wet and slick.
- **Long sleeves / appropriate clothing** — no exposed skin where splashes are likely.

⚠ **HEADS-UP**
Respiratory protection is required when ventilation is inadequate or when working over fuming tanks (acid pickle, the air-agitated copper bath, HCl-based activation). Use the cartridge or supplied-air respirator your shop specifies. If you can smell acid fumes strongly, the ventilation is not keeping up — stop and report it.

💡 **TIP**
Put your gear on *in order*: boots and apron first, then gloves, then goggles, then face shield last. Take it off in reverse. This keeps you from touching your face with contaminated gloves. And rinse your gloves *before* you take them off.

STATION-BY-STATION HAZARD SUMMARY

STAGE	PRIMARY HAZARD	WHY IT'S DANGEROUS
01 Soak / Electroclean	**Hot alkaline (caustic)**, 130–180°F; electrolytic gas	Chemical burns; hot splashes; H_2/O_2 gas evolution
02 Rinse	Residual alkaline cleaner	Mild skin/eye irritant; wet/slip hazard
03 Acid Activation	**Strong mineral acid**; HCl **fumes**; flammable **H_2 gas**	Severe burns & eye damage; corrosive fumes; fire risk
04 Copper Strike (cyanide)	**CYANIDE — lethal.** Acid contact releases **HCN gas**	**Can be fatal.** Strict acid segregation; alkaline burns
04 Strike (non-CN / Woods Ni)	Alkaline (non-CN) or strongly acidic + **nickel** (Woods)	Burns; nickel is a skin sensitizer / suspect carcinogen
05 Acid Copper Plating	**Sulfuric acid** bath; **acid / copper-sulfate mist**; high-amperage rectifier	Burns; respiratory irritation; shock & arc-flash; aquatic toxicity
06 Rinse	Dilute acidic copper residue	Irritant; aquatic toxicity; slip hazard
07 Post-Treatment / QC	Varies: anti-tarnish (Se/sulfide), lacquer solvent, or **hex chromate**	Depends on path; Cr(VI) is a **carcinogen**; solvents flammable
08 Final Rinse/Dry	Hot water/air; residual chemistry	Burns from hot dryer; slip hazard

⚠ **HEADS-UP — EXPOSURE LIMITS (VERIFY BEFORE PUBLISHING/RELYING)**
Cited occupational exposure limits carried from the source content — **confirm against current OSHA tables and your product SDS:** sulfuric acid mist **1 mg/m³** `[VERIFY]`; copper dusts/mists **1 mg/m³**, copper fume **0.1 mg/m³** as Cu `[VERIFY]`; soluble nickel **1.0 mg/m³** as Ni `[VERIFY]`; hexavalent chromium **5 µg/m³** `[VERIFY]`; cyanide (as CN) `[VERIFY]`.

⚠ **HEADS-UP**
Never eat, drink, smoke, vape, or apply cosmetics on the line. Hand-to-mouth contact is a major route for ingesting plating chemicals — especially heavy metals like copper and nickel, and (where present) cyanide and chromium. Wash thoroughly before breaks and before leaving. Keep food and drink completely out of the work area.

CHAPTER 6

EMERGENCY & FIRST-RESPONSE

WHEN SECONDS COUNT — LEARN THIS COLD, BEFORE YOU EVER NEED IT

Know these *before* you need them. In an emergency, seconds matter and you won't have time to read. **Locate, before your first shift on the line:** the nearest **emergency eyewash** (within ~10 seconds' reach of every chemical station), the nearest **safety shower**, the **fire extinguisher** and alarm, the **rectifier emergency stop / disconnect** and main power shutoff, the **spill kit** and **SDS binder** (Safety Data Sheets), and — if a cyanide strike is on the line — the **cyanide antidote kit** and posted cyanide emergency procedure.

> ⚠ **HEADS-UP**
> **Eyewash and shower stations must always be unblocked.** Never stack parts, totes, or drums in front of them. The day you need one is the day you can't move boxes out of the way.

FIRST RESPONSE BY HAZARD

SITUATION	WHAT TO DO — IMMEDIATELY
Chemical on skin	Get to the safety shower; flush **at least 15 minutes**. Remove contaminated clothing while flushing — don't stop early because "it feels better." Get medical attention; bring the SDS.
Chemical in eyes	Go to the eyewash; hold eyelids open and flush **at least 15 minutes**, rolling your eyes. Do not rub. Get medical attention immediately — eye exposures are always urgent. Bring the SDS.
Suspected cyanide exposure / HCN gas	**Evacuate to fresh air immediately** and alert others — HCN can incapacitate in seconds. Call emergency medical response now. Trained responders administer the **cyanide antidote per your facility program** `[VERIFY protocol]`. Do not enter a suspected HCN area without proper respiratory protection.
Inhaled fumes/mist, feel ill	Move to fresh air immediately. Report it — do not "tough it out." Seek medical attention for anything beyond brief, mild irritation. Bring the SDS.
Chemical swallowed	**Do not** induce vomiting unless the SDS directs it. Follow SDS first aid; call medical help / poison control immediately.
Electrical shock / arc-flash	**Do not touch the person** if they may still be in contact with current. Cut power at the emergency stop / disconnect first, then call for help and render aid.
Chemical spill	Alert others and keep people away. Only clean it if trained and within your shop's "small spill" limits — otherwise evacuate and call trained spill response. **Never let acid and cyanide wastes mix**, and never let spills reach floor drains.

> ⚠ **HEADS-UP — THE CYANIDE RULE (LIFE OR DEATH)**
> If your line uses a cyanide strike: **acid and cyanide must NEVER meet.** Acid contacting a cyanide bath, cyanide-wet parts, or cyanide-contaminated rinses releases **hydrogen cyanide (HCN) gas — potentially fatal.** Dedicated tanks, dedicated rinses, and segregated cyanide waste are mandatory. Know where the antidote kit is and who is trained to use it, before your first shift near that tank.

> ⚠ **HEADS-UP — LOCKOUT/TAGOUT (LOTO)**
> LOTO is mandatory for *all* rectifier and electrical maintenance. LOTO means physically locking the power off and tagging it so no one can switch it back on while someone is working on it. "I'll just be a second" is how people get electrocuted.

> ⚠ **HEADS-UP — A CRITICAL ACID-MIXING RULE, FOR LIFE**
> When diluting acid, **always add ACID to WATER — never water to acid.** Adding water to concentrated acid releases heat so violently it can boil and erupt the acid into your face. The memory aid: "**Do as you oughta — add acid to water.**" Pour slowly, down the side, with the acid going *into* a larger volume of water.

CHAPTER 7

THE SAFETY PHILOSOPHY OF AN ACID COPPER LINE

THE MINDSET THAT KEEPS YOU SAFE ACROSS A WHOLE CAREER

Rules and PPE are the *tools*. The philosophy is the *mindset* that makes them work. Five principles guide everything on this line.

1. ASSUME EVERY TANK CAN HURT YOU.

Even the rinse tanks carry residual chemistry. Even “just water” can be slippery enough to put you on the floor. Respect comes first; familiarity is where complacency — and accidents — begin. The operator who has run the line for ten years without an incident got there by *staying* careful, not by relaxing.

2. CONTAINMENT BEATS CLEANUP.

Move parts deliberately. Let them drain over the tank before transferring. Don’t fling racks, splash, or rush handoffs. Most chemical exposures aren’t dramatic spills — they’re splashes and drips from sloppy handling. Slow, smooth movement is safer *and* makes better parts (less drag-out means cleaner rinses and less cross-contamination).

3. RESPECT THE TWO FACES OF EVERY HAZARD.

On this line, the things that make good parts are the same things that can hurt you. The acid that activates the metal burns your skin. The current that deposits copper can arc-flash. The strike that saves your adhesion may be lethal cyanide. You can’t separate the work from the hazard — so you manage both at once, every time.

4. THE CHEMISTRY IS UNFORGIVING, SO BE DISCIPLINED.

Acid copper punishes contamination — particulate, chloride drift, chromium drag-back — and the strike punishes carelessness even harder. The same discipline that keeps the bath healthy (careful rinsing, no shortcuts, controlling drag-in and drag-out) is the discipline that keeps *you* healthy. Good housekeeping is a safety practice, not just a quality practice.

5. WHEN IN DOUBT, STOP AND ASK.

No one expects you to have every answer, especially while learning. The unsafe operator is not the one who asks questions — it’s the one who guesses. If a fume seems strong, a glove looks worn, a reading looks wrong, or a procedure feels off, *stop and check.* That instinct is the single most valuable safety habit you can build.

THE HAZARD FAMILIES TO INTERNALIZE

1. **Acids & Alkali (corrosives).** The hot caustic cleaner (Stage 1), the mineral acids (Stage 3), the strongly alkaline strikes, and the sulfuric-acid plating bath all cause burns on contact and damage eyes instantly. *Defense:* goggles, face shield, gloves, apron — the 15-minute flush rule if exposed — and always acid-to-water.
2. **Electrical & physical.** The rectifier runs high amperage — shock and arc-flash risk. Wet floors mean slips. *Defense:* LOTO for all electrical work, never touch a rectifier or bus bar with wet hands, non-slip boots, keep walkways clear.
3. **Copper-sulfate & mist / metals.** Copper sulfate is an irritant and highly toxic to aquatic life; air agitation makes acid/copper mist; nickel (Woods strike) is a sensitizer/suspect carcinogen. *Defense:* ventilation and mist control, respirator where needed, waste segregation, gloves always.
4. **Cyanide (the family that can kill in seconds).** Where a cyanide strike is used, acid contact makes lethal HCN. *Defense:* absolute acid/cyanide segregation, dedicated rinses, segregated waste, trained response, antidote kit on hand.

⚠ **HEADS-UP — THE CYANIDE RULE (THE SINGLE MOST IMPORTANT LINE IN THIS MANUAL)**

Both strike chemistries this manual treats as **co-equal** options — cyanide and alkaline non-cyanide — but the cyanide one carries a rule that overrides everything else: **NEVER allow acid to contact a cyanide bath, cyanide-wet parts, or cyanide-contaminated rinses. Acid + cyanide = hydrogen cyanide gas = potentially fatal.** If your shop runs the non-cyanide strike instead, you avoid this hazard entirely — but you must still know the rule, because you may work in a shop that runs cyanide.

★ **REMEMBER**

You’ve now got the whole foundation: what plating is, what makes acid copper special, why active metals need a strike, how the eight-stage line works as one connected system, why the order is locked by chemistry, what to wear, what to do in an emergency, and the safety mindset that ties it all together. Everything in the stage chapters builds on these fundamentals. Turn the page knowing the *why* — and the *how* will make a lot more sense.

PART TWO — THE PREP LINE (STATIONS 01–04)
STATION 01 OF 08

SOAK CLEAN / ELECTROCLEAN

"COPPER LEVELS SCRATCHES BEAUTIFULLY. IT LEVELS NOTHING YOU FAILED TO CLEAN OFF."

★ **REMEMBER — THE PREP-LINE PROMISE**

By the time a part reaches the copper tank, most of whether it'll turn out beautiful or scrap has already been decided. The four prep stations — Clean, Rinse, Activate, Strike — are not the "boring stuff before the real plating." As the old-timers say: *"The plating bath gets the credit, but the pretreatment does the work."*

Welcome to the very first tank. Everything good that happens downstream starts here. Your job at this station is simple to say and important to nail: **get the part perfectly clean.** Not "looks clean." Not "pretty clean." *Chemically* clean — no oil, no grease, no fingerprints, no shop dust. If you do this one thing right, you've made every job after you easier. If you cut a corner here, that corner comes back as a pit, a skip, or a peel that no amount of bright leveler downstream can hide.

WHAT YOU'RE DOING & WHY IT MATTERS

When parts arrive at your station, they are *not* clean — even if they look shiny. They almost always carry three layers of stuff:

1. **Particulate** — loose solid bits: shop dust, metal fines from machining, grit.
2. **Oil and grease** — the slippery film from stamping oils, cutting fluids, rust-preventives, and hand contact. You usually can't see it, but it's there.
3. **Oxide scale** — a thin layer of oxide/tarnish bonded to the metal itself. (We don't remove this one here — that's Station 03's job. But it's good to know it's there.)

Soak Clean removes the first two. The cleaner is a hot, strongly alkaline water-based solution. It works several ways at once: **heat** softens and thins the oils so they let go of the metal; **surfactants** (the soap-like ingredients) surround tiny droplets of oil and lift them off — exactly like dish soap cutting grease off a frying pan; and **builders / alkalinity** chemically break down (saponify) certain oils, turning them into something water can carry away. For steel and zinc die cast headed for a strike, an **electroclean** step after the soak is strongly recommended — passing current through the part in the cleaner generates scrubbing gas that strips the last microscopic film.

💡 **TIP — THE WATER BREAK TEST (FREE, FAST, AND NEVER LIES)**

A truly clean surface loves water. When you pull a clean part out, the water clings and **sheets off in one continuous, unbroken film.** A dirty surface repels water, so the water **beads up** into little droplets instead. **Continuous sheet = PASS** (send it on). **Beading or broken film = FAIL** (clean it again). The test costs nothing and takes two seconds. Use it on every rack, every time.

```
FAIL - water beads up         PASS - water sheets evenly
+--------------+              +--------------+
|  o   o   o   |  X           |##############|  OK
|    o    o    |              |##############|
+--------------+              +--------------+
Oil remains -> RE-CLEAN       Surface clean -> proceed
```

⚙ **TECHNICAL STUFF**

"Water break" refers to the *surface energy* of the metal. Clean metal has high surface energy, so water spreads to maximize contact (low contact angle). Even a one-molecule-thick film of oil drops the surface energy, so water pulls itself into beads to minimize contact (high contact angle). That's why the test can catch contamination far too thin for your eye to see.

OPERATOR ESSENTIALS

MEMORIZE THE RANGES

PARAMETER	RANGE	WHAT TO WATCH
Soak temperature	130–180°F (54–82°C)	**Do not exceed ~190–200°F**; milder for zinc die cast
Soak time	3–10 min	Heavy draw oil: 8–10 min · Light stamping oil: 3–5 min
Concentration	4–8 oz/gal (30–60 g/L)	Titrate (test) weekly at minimum
Electroclean polarity	Cathodic → Anodic (reverse last 15–30 s)	Finish **anodic** to avoid smut redeposition
Electroclean V / CD	4–8 V; 30–80 ASF	Don't over-cathodic on high-strength steel
Electroclean time	1–3 min	Limit cathodic time on zinc die cast (blistering)
pH	11.0–13.5	Depends on the cleaner formulation
Water break test	Pass = continuous sheet	The only reliable field test

A few plain-language definitions: **Concentration** = how much cleaner chemical is dissolved in the water; too weak and it won't cut the oil, too strong wastes money. **Titrate** = a simple test that measures the actual concentration so you know when to add more. **Cathodic vs. anodic** = which way the current flows through the part: *cathodic* (part negative) makes about twice the scrubbing gas but can plate metallic *smut* onto the part, so you finish *anodic* (part positive) for the last 15–30 seconds to strip that smut off. On zinc die cast, keep cathodic time short to prevent hydrogen blistering of the casting.

⚠ **HEADS-UP — SAFETY: HOT ALKALINE SOLUTION & ELECTROLYTIC GAS**

This tank is **hot AND caustic.** Splashes cause **chemical burns on contact**, and the steam/mist irritates your airways. Electrocleaning also releases **hydrogen and oxygen gas** — ventilation required; no open flames near the tank; the tank carries DC current, so **LOTO before reaching in.**

PPE every time: chemical splash goggles, face shield (when charging or dumping the tank), elbow-length chemical gloves, apron, chemical-resistant boots. First aid: **skin contact → flush 15 minutes; eye contact → eyewash 15 minutes** and get help.

STEP-BY-STEP PROCEDURE

1. **Check the tank before you start.** Confirm temperature is in range (never above ~190–200°F). Confirm concentration is in range (check the last titration log). Make sure the skimmer is running and the surface is reasonably free of floating oil.
2. **Inspect the load.** Heavy rust-preventive oil → the long end of the time range (8–10 min). Light stamping oil → 3–5 min. Note the substrate — zinc die cast needs a milder, cooler cleaner and shorter cathodic time.
3. **Rack or load the parts** so solution reaches every surface. Avoid nesting where air can be trapped (trapped air = a dry spot = a dirty spot). For tubular/blind-hole parts, position them so the bore floods and drains.
4. **Soak fully** for the timed interval with agitation running.
5. **If electrocleaning:** run cathodic, then **reverse to anodic for the last 15–30 seconds** to remove smut. Keep cathodic time short on zinc die cast.
6. **Withdraw slowly,** letting solution drain back into the tank (saves chemical, reduces drag-out).
7. **Run the water break test.** Continuous sheet → Station 02. Beading → back in the cleaner.
8. **Move the rack promptly** so the clean surface doesn't sit and re-oxidize, and log it if your shop logs cleaning.

TEACH-BACK CHECKLIST

A trainee is signed off on Station 01 only when they can, unprompted:

- ☐ Name the two kinds of contamination the soak clean removes (particulate and oil/grease — it does NOT remove oxide scale; that's Station 03).
- ☐ State the soak temperature range and the ceiling (**130–180°F; do not exceed ~190–200°F**).
- ☐ State the time for heavy oil vs. light oil (8–10 min vs. 3–5 min).
- ☐ Explain why electrocleaning finishes **anodic** (to strip metallic smut before it rides forward).
- ☐ Correctly perform a water-break test and read the result honestly (continuous sheet = pass).
- ☐ Explain what they'd do if the water-break test fails (re-clean it — never send it forward), and name three pieces of required PPE.

TOP MISTAKES NEW OPERATORS MAKE

1. **Skipping the water break test because the part "looks fine."** Oil that ruins plating is invisible. Test every time.
2. **Running the tank too cold.** Cold cleaner can't cut oil no matter how long you soak. If the water break keeps failing, check temperature first.
3. **Ending electroclean cathodic.** You leave metallic smut on the part that wrecks strike adhesion. Always reverse to anodic to finish.
4. **Pulling parts through the floating oil layer.** If the skimmer is off or overwhelmed, the oil you just removed redeposits on the way out.
5. **Over-cathodic on zinc die cast.** Too much cathodic time hydrogen-blisters the casting. Keep it short and milder.
6. **Nesting parts so tight that air pockets form.** Wherever solution can't reach, the part stays dirty. Rack with space.

QUICK TROUBLESHOOTING

SYMPTOM	PROBABLE CAUSE	WHAT TO DO
Water break fails	Low concentration or temp	Titrate cleaner; bring temp to range
Oily film after cleaning	Floating oil redepositing; exhausted cleaner	Skim/run skimmer; dump tank if heavily contaminated
White residue on parts	Hard-water salts; cleaner residue	Check rinse-water quality (use DI)
Etching / discoloration of zinc die cast	Cleaner too alkaline/hot for substrate	Reduce temp; use die-cast-safe cleaner
Skip plating (bare spots) downstream	Passive film / fingerprints not removed	Finish electroclean anodic; clean gloves only

SIGN-OFF — STATION 01

Trainee: ______________________ Trainer: ______________________ Date: ____________

"I confirm the parts in this load passed the water break test (continuous sheet, no beading), the soak clean was in temperature and concentration, electrocleaning (where used) finished anodic, and I wore required PPE."

STATION 02 OF 08

RINSE (PRE-ACTIVATION)

"A RINSE ISN'T A REST STOP. IT'S A CHECKPOINT."

You just got the part beautifully clean — but it's now coated in a thin film of the cleaner itself. This station's whole job is to **wash that cleaner film off** before the part goes into the acid tank. It sounds like a "nothing" step. It isn't: a weak rinse here quietly sabotages the acid tank next door, carries silicates and surfactants downstream, and can eventually foul the strike bath — and you won't see the damage until parts come out wrong.

• WHAT YOU'RE DOING & WHY IT MATTERS

The soak cleaner is **alkaline** (high pH). The very next tank — acid activation — is, well, **acid** (low pH). Acid and alkaline are chemical opposites: when they meet, they neutralize each other and cancel out. So if you carry a film of alkaline cleaner straight into the acid tank, three bad things happen:

1. **You waste acid** — the acid gets used up neutralizing the cleaner instead of doing its real job on the part.
2. **You raise the acid tank's pH** — making it weaker and less effective at activating the surface.
3. **You carry cleaner chemistry down the chain** — silicates and surfactants that show up later as poor adhesion and can foul the strike.

This invisible carry-over of liquid from one tank to the next is called **drag-out** (what leaves the tank clinging to the part) or **drag-in** (what arrives in the next tank). The rinse station's job is to knock that drag-out down to almost nothing — by dilution. Clean water surrounds the part and the cleaner film dissolves off into the rinse water, where it's diluted to harmlessness and flushed away by the constant flow of fresh water.

> 💡 **TIP — HOW WE MEASURE A CLEAN RINSE: CONDUCTIVITY**
> You can't see dissolved cleaner, so we measure it with **conductivity** — how well the water conducts electricity. Pure water barely conducts at all. Dissolved chemicals make water conduct *more*. So **higher conductivity = more cleaner left in the rinse = dirtier rinse.** We measure it in **microsiemens per centimeter (µS/cm)** — just think of it as "the contamination number." Lower is cleaner.

> ★ **REMEMBER — THE RULE CARD NUMBER: < 500 MS/CM**
> Keep this rinse **below 500 µS/cm** (aim for **under 300**). Above 500, you're dragging alkaline cleaner into the acid activation tank — wasting acid, raising its pH, and carrying cleaner chemistry toward the strike. This number is your early-warning system. Check it daily. On copper lines feeding PCB work, use DI water and hold it tighter still.

• THE SMART WAY TO RINSE: COUNTER-FLOW

You *could* just blast fresh water through a single tank and dump it. It works, but wastes enormous water. The professional approach is a **counter-flow (counter-current) rinse**: the parts move forward; the water flows backward. Fresh water enters the *last* (cleanest) tank, then overflows backward into the dirtier tank. So your part gets progressively cleaner rinses, and the cleanest water it ever sees is the last thing to touch it before the acid tank.

```
FROM        +-------------+  PARTS ->  +-------------+        TO
CLEANER ->  |  RINSE 1    | ---------> |  RINSE 2    | ---->  ACID
            |  (dirty)    |            |  (clean)    |
            | <500 uS/cm  |            | aim <300    |
            +-------------+            +-------------+
                 ^   overflow <----------    ^
                 +------ <- FRESH WATER FLOWS BACKWARD <- -----+
                                                       FRESH IN
```

OPERATOR ESSENTIALS

PARAMETER	RANGE	NOTES
Temperature	`Ambient (60–85°F / 16–29°C)`	No heating required
Time	`30–60 sec with agitation`	Longer for blind holes/tubing
Water source	`Municipal or DI`	**DI preferred** for copper lines feeding PCB work
Conductivity	`< 500 µS/cm (target < 300)`	Monitor daily
Flow rate	`2–4 gal/min`	Continuous overflow or counter-current
Tank turnover	`4–6 / hour`	Prevents stagnation
pH	`7–10`	Elevated pH = cleaner drag-in

Plain-language definitions: **Ambient** = room temperature, no heater. **DI water** = deionized water with minerals removed (low conductivity, rinses better, no spots). **Continuous overflow** = fresh water in, dirty spills out the top. **Tank turnover** = how many times per hour the whole tank volume is replaced; 4–6 keeps it from going stale.

> ⚠ **HEADS-UP — SAFETY: RESIDUAL ALKALINE CLEANER**
> The rinse water carries dragged-in cleaner and starts out mildly caustic. Wear **goggles, chemical gloves, and non-slip footwear** — the floor around rinses is always wet, and slips are the #1 injury here.

STEP-BY-STEP PROCEDURE

1. **Check conductivity at shift start.** If it's near or above 500 µS/cm, tell your lead — the rinse needs more flow or a refresh.
2. **Confirm water is flowing** (continuous overflow, or counter-current cascade active).
3. **Transfer the rack promptly** from the cleaner — dried cleaner is harder to rinse off.
4. **Immerse fully with agitation, 30–60 seconds.** For blind holes and tubing, go longer and move the part to flush trapped solution.
5. **Lift and drain** over the rinse tank for a few seconds to carry less forward.
6. **Move promptly to the acid tank** — don't let the rinsed part sit and re-oxidize.

TEACH-BACK CHECKLIST

- [ ] Explain why carrying cleaner into the acid tank causes problems (neutralizes acid — wastes acid, raises pH, carries cleaner chemistry toward the strike).
- [ ] Name the term for chemistry clinging to a part from the previous tank (drag-out / drag-in).
- [ ] Explain what conductivity tells us and which direction is "clean" (lower = cleaner).
- [ ] State the conductivity limit and target (**< 500 µS/cm; target < 300**).
- [ ] State how long to rinse and why longer for tubing (30–60 sec; tubing traps solution).

TOP MISTAKES NEW OPERATORS MAKE

1. **Treating the rinse as "quick."** A two-second dunk rinses nothing — give it the full 30–60 seconds.
2. **Ignoring the conductivity meter.** A high number means you're weakening the acid tank right now.
3. **Letting the rack drip-dry between cleaner and rinse.** Dried-on cleaner is stubborn — move promptly.
4. **Not flushing blind holes and tubes,** so concentrated cleaner hides in cavities and rides forward.
5. **Skipping the drain step** — a few seconds of draining sharply cuts what you drag into the acid.

QUICK TROUBLESHOOTING

SYMPTOM	PROBABLE CAUSE	WHAT TO DO
High conductivity	Heavy drag-out; low flow	More drain time off cleaner; more rinse flow; add counter-current stage
Soapy residue on parts	Inadequate rinse time	Extend dwell; add agitation; double rinse
Acid tank going alkaline	Cleaner carry-over	Improve this rinse; add a second stage
Foaming in the rinse	Surfactant from cleaner	Increase water flow; check cleaner

SIGN-OFF — STATION 02

Trainee: ____________________ Trainer: ____________________ Conductivity: __________ µS/cm

"I confirm the pre-activation rinse conductivity was within limit (< 500 µS/cm), water flow was running, and parts were rinsed the full time with PPE worn."

STATION 03 OF 08

ACID ACTIVATION

“OXIDE IS INVISIBLE. THE BLISTER IT CAUSES THREE TANKS LATER IS NOT.”

This is the station that makes the strike (and everything after it) *stick*. Your clean part still has an invisible, microscopic layer of oxide bonded to the surface. Copper — and the strike that goes on first — can only bond to bare, chemically “awake” metal. The acid dip dissolves that oxide and leaves a fresh, reactive surface. This is also the last acid tank before the strike, so it carries a special safety weight: **if the next tank is a cyanide strike, acid drag-out from here must never reach it.**

• WHAT YOU’RE DOING & WHY IT MATTERS

Even a spotlessly cleaned part has a thin film of **oxide** on its surface — at its worst visible rust or tarnish, at its mildest an invisible film that forms in *minutes* on bare metal exposed to air. Either way, it’s a barrier. A strike (or copper) plated on top of oxide doesn’t truly bond — it sits on the oxide like paint on a greasy wall, and it eventually flakes or blisters.

Acid activation (also called the **acid dip** or **pickle**) fixes this. The acid chemically dissolves the oxide layer right off the surface, leaving **clean, chemically active metal** ready to form a strong bond. On steel you may see tiny bubbles — that’s **hydrogen gas (H_2)**, a normal sign the acid is reacting with the metal (and a clue to a hidden hazard below).

> ★ **REMEMBER — THE RULE CARD: 5-60 SECONDS**
>
> Dip time depends entirely on the substrate. **Under-dip** = oxide remains, adhesion fails downstream. **Over-dip** = you etch and pit the part, load steel with hydrogen, and (on brass) *dezincify* the surface into copper smut. Steel wants 15–60 s; brass, copper alloys, and zinc die cast want just 5–15 s. Match the metal, and time it.

> ⚙ **TECHNICAL STUFF**
>
> The reaction is straightforward: the acid reacts with the surface oxide to dissolve it (metal oxide + acid → soluble salt + water), and once the oxide is gone the acid begins attacking the base metal itself, producing hydrogen gas on steel ($Fe + 2HCl \rightarrow FeCl_2 + H_2\uparrow$). That last reaction is why you must *time the dip* — a little attack cleans and activates; too much etches the part and floods steel with hydrogen. On brass, over-attack selectively removes zinc (**dezincification**), leaving a weak, coppery smut.

> ⚠ **HEADS-UP — THE FIREWALL BEFORE THE STRIKE**
>
> This is the last acid the part sees before the strike. If the strike is **cyanide**, acid dragged forward from here can release **lethal HCN gas** in that tank. Drain thoroughly and rinse (per your shop’s layout) before the strike. **Acid and cyanide must never meet.**

OPERATOR ESSENTIALS

PARAMETER	RANGE	NOTES
Temperature	Ambient–110°F (Amb–43°C)	Ambient typical
Time — steel/iron	15–60 sec	Light oxide 15–30 s; heavier 30–60 s
Time — brass / copper	5–15 sec	Short dip — acid attacks copper alloys fast; dezincifies brass
Time — zinc die cast	5–15 sec	Minimal; base metal dissolves fast; use mild/proprietary activator
Acid — Option A	H_2SO_4 5–15% v/v	**Preferred on copper lines** — adds no chloride; low fuming
Acid — Option B	HCl 10–30% v/v	Good oxide removal; fumes — ventilation required; adds chloride (mind drag-in)
Agitation	Manual rack movement only	**No air agitation** — it creates acid mist
Iron content	Dump when spent (>~50 g/L Fe)	Spent acid loses bite; drags metals forward

Plain-language definitions: **H_2SO_4** = sulfuric acid; **HCl** = hydrochloric (muriatic) acid. Both are strong mineral acids. **% v/v** = “percent by volume” — how much concentrated acid is mixed into the water. On a copper line, sulfuric activation (Option A) is often preferred because it *doesn’t add chloride* to the chain — and chloride control is precious in the copper bath downstream. HCl removes oxide aggressively but fumes and adds chloride you then have to account for.

SUBSTRATE-SPECIFIC NOTES

- **Steel / iron:** Standard H_2SO_4 or HCl activation. Use *inhibited* acid on >40 HRC steel to limit hydrogen absorption. Steel goes to a **strike**, never directly to acid copper.
- **Brass / copper alloys:** Short dip only (5–15 s). Excess time dezincifies brass and leaves copper smut. Dilute H_2SO_4 is gentler. These may proceed to direct acid copper after excellent activation, but a strike improves adhesion.
- **Zinc die cast:** Extremely sensitive — mild/proprietary activator, minimal time. Must go to a strike (cyanide or alkaline non-cyanide copper).
- **Aluminum:** Requires a **zincate** (zinc immersion) pretreatment, then a strike — not covered by this simple acid dip.

⚠ **HEADS-UP — SAFETY: STRONG MINERAL ACID (READ THIS TWICE)**

HCl fumes at room temperature — you don’t have to heat it to release irritating, corrosive vapor; ventilation is required. **H_2SO_4 causes severe dehydration burns** and reacts violently with water. (That’s exactly why you dilute by trickling a *small* amount of acid into a *large* volume of water.) **Hydrogen gas released during pickling is flammable** — no open flames or sparks near the tank.

PPE: goggles + face shield, acid-resistant gloves, apron, boots, **respirator if ventilation is inadequate.** Skin/eye contact: flush 15 min. **Spills:** neutralize with soda ash. **Always add acid to water. No air agitation** — move the rack by hand.

THE BIG ONE: HYDROGEN EMBRITTLEMENT

⚠ **HEADS-UP — SAFETY & QUALITY: HYDROGEN EMBRITTLEMENT**

This is the most important concept in your whole section, so we'll go slow. When acid attacks steel, it releases hydrogen. Most of it bubbles away as gas — but some sneaks *into* the steel as single hydrogen atoms, wedging between the iron atoms. On ordinary mild steel this is usually harmless. But on **high-strength / hardened steel**, that trapped hydrogen makes the metal **brittle** — and a brittle part can **crack and fail suddenly under load**, sometimes hours or days later. On a critical fastener or structural part, that's a real safety failure.

The acid pickle is a major source of hydrogen absorption on the line. Uninhibited acid on high-strength steel (**>40 HRC / >180 ksi UTS**) can load the lattice with hydrogen quickly.

What you do about it: Use the **minimum effective acid concentration and dip time** — don't over-dip "to be safe." **Always use an inhibitor on substrates over 40 HRC.** If a job is flagged high-strength, **stop and check with your lead** — hardened parts require a post-plate **relief bake** (handled at Station 07) to drive hydrogen back out.

Terms: **HRC** = Rockwell C hardness (higher = harder). **ksi UTS** = thousands of psi of ultimate tensile strength. Treat these words on a job traveler as a "slow down and check" flag.

STEP-BY-STEP PROCEDURE

1. **Check the job for hardness flags first.** If the traveler says high-strength, hardened, >40 HRC, or >180 ksi — pause and confirm the right process and that inhibitor is in use.
2. **Confirm ventilation is running** and your PPE (including face shield) is on.
3. **Verify acid type/strength and the dip time** for this substrate: 15–60 s steel; 5–15 s brass/copper/zinc die cast.
4. **Immerse fully** and start your timer. **Agitate by gently moving the rack by hand** — never air.
5. **Watch the surface.** The dull oxide should visibly lighten. On brass, pull promptly — a coppery smut means you've gone too long (dezincification).
6. **Pull at the timed mark.** Don't "round up."
7. **Drain thoroughly** and move **promptly** to the strike (Station 04). If the strike is cyanide, confirm draining/rinsing per your shop's procedure — acid must not reach the cyanide tank.
8. **Watch the iron level over time.** When the bath is spent (>~50 g/L Fe, or your shop's limit), flag it for dumping/recharge.

TEACH-BACK & QUICK TROUBLESHOOTING

SYMPTOM	PROBABLE CAUSE	WHAT TO DO
Dark oxide remains on steel	Acid too dilute; time too short	Increase concentration; extend time *(check HE risk first)*
Pitting / etching of substrate	Too concentrated/long/warm	Reduce concentration; shorten dip; add inhibitor
Copper smut on brass	Dezincification (over-dipped)	Shorten dip; go to a copper strike
Adhesion failure downstream	Incomplete activation; residual oxide; alkaline drag-in	Re-check acid strength/time; tighten Station 02 rinse
Excessive HCl fuming	High conc/temp; poor ventilation	Reduce to minimum effective; check exhaust
HE cracking/failures	Over-long pickle; no inhibitor; >40 HRC steel	Shorten pickle; add inhibitor; review hardness; relief bake

Teach-Back: Explain what the acid dip removes and why; the dip time for steel vs. brass/zinc die cast; why there is no air agitation; hydrogen embrittlement and which parts are at risk (>40 HRC); why acid must never reach a cyanide strike; and the rule for diluting acid (acid to water).

SIGN-OFF — STATION 03

Trainee: ____________________ Trainer: ____________________ Dip used: ________ sec · Inhibitor? Y / N

"I confirm I checked the job for hardness flags, ran the acid dip at the timed interval for the substrate with manual agitation only, ventilation and PPE in place, drained so no acid reaches a cyanide strike, and used inhibitor where required."

STATION 04 OF 08 · THE MAKE-OR-BREAK LAYER

COPPER STRIKE

THE THIN ADHERENT FLASH THAT STOPS IMMERSION COPPER ON ACTIVE METALS

This station has no equivalent on a simple zinc line, and it is the reason acid copper works at all on steel. Your job here is to lay down a **thin, tightly-bonded flash of copper (or nickel)** under controlled conditions — so that by the time the part reaches the acid copper tank, it presents a compatible, noble surface for the copper to build on. Skip it or botch it on an active metal, and everything downstream blisters. As the tagline goes: *“Acid copper on bare steel is a chemistry experiment in how to grow blisters. Strike first.”*

• WHAT YOU’RE DOING & WHY IT MATTERS

Copper is **more noble** (less reactive) than steel, zinc, and aluminum. Put any of those metals into an acid copper bath and copper deposits on them **by immersion — a purely chemical swap, before a single amp flows.** That immersion copper is loose, powdery, and non-adherent. Plate over it and it lifts, taking your whole deposit with it as blisters.

★ **REMEMBER — WHY THE STRIKE EXISTS**

$Cu^{2+} + Fe^{0} \rightarrow Cu^{0} + Fe^{2+}$. In acid copper, copper ions rip electrons off the steel and slap themselves on as loose, non-adherent copper — while iron dissolves into the bath. The strike lays down a thin, *tightly bonded* copper (or nickel) layer under conditions where immersion doesn’t happen, giving the acid copper a noble surface to grow on. On steel, zinc die cast, and aluminum the strike is **mandatory**.

On copper and brass substrates — already noble enough — a strike may be omitted after thorough activation and a well-managed guard rinse, but it still improves adhesion and is recommended for critical work. The strike layer also does a second job: together with the rinse that follows it, it serves as the **pre-plate contamination barrier** for the sensitive acid copper bath.

💡 **TIP**

Two words decide this station: **complete** and **live.** The flash must *completely* cover the part before it enters acid copper (any bare spot will immersion-plate), and you enter the strike *live* (current already on) so no brief immersion deposit forms at the moment of insertion. We’ll drill both.

⚠ **HEADS-UP — READ BEFORE YOU TOUCH THIS TANK**

The traditional copper strike is a **cyanide** bath. Cyanide is a **lethal poison**, and **acid contacting cyanide releases hydrogen cyanide (HCN) gas that can be fatal.** Many modern shops run an **alkaline non-cyanide** strike instead to avoid this hazard entirely. This manual treats the two as **co-equal** options — but whichever your shop runs, you must understand the cyanide rule before working near a cyanide tank. Full details on the next pages.

• YOUR STRIKE OPTIONS (CYANIDE AND NON-CYANIDE ARE CO-EQUAL)

There is no single “right” strike — there are options, each with tradeoffs. The two mainstream choices for steel and zinc die cast are the **cyanide** strike and the **alkaline non-cyanide** strike; this manual presents them as **equally valid**. Woods nickel and copper pyrophosphate are specialty strikes for particular substrates.

STRIKE TYPE	CHEMISTRY (TYPICAL)	BEST FOR	NOTES / TRADEOFFS
Cyanide copper strike	CuCN 15–30 g/L; free NaCN/KCN; carbonate; often Rochelle salt	Steel, zinc die cast	Excellent adhesion & coverage; **cyanide hazard** — segregate, never contact acid
Alkaline non-cyanide copper strike	Proprietary alkaline (pyrophosphate / complexed) copper	Steel, zinc — cyanide-free shops	RoHS/cyanide-free; adhesion approaching cyanide with good control; per TDS
Woods nickel strike	$NiCl_2$ ~240 g/L + HCl ~125 mL/L	Stainless, difficult steels; nickel barrier	Activates passive surfaces; strongly acidic; nickel is a sensitizer
Copper pyrophosphate strike	Cu pyrophosphate, near-neutral	Electronics; sensitive substrates	Near-neutral; gentle; specialty

• OPERATOR ESSENTIALS (REPRESENTATIVE — FOLLOW SUPPLIER TDS)

PARAMETER	CYANIDE CU STRIKE	ALKALINE NON-CN CU	WOODS NICKEL
Temperature	100–140°F	100–130°F	Ambient–90°F
pH / nature	Strongly alkaline	Alkaline	Strongly acidic (HCl)
Current density	10–40 ASF	Per TDS	20–60 ASF
Time	1–5 min (thin flash)	1–5 min	1–4 min
Anodes	Copper (bagged)	Copper	Nickel
Deposit	~2.5–5 µm flash	Thin flash	Thin flash

★ **REMEMBER**

The strike is a **flash, not a build** — roughly **0.0001–0.0002 in (2.5–5 µm)**. Its job is coverage and adhesion, not thickness. The *thickness* comes later, in the acid copper tank. Leave a strike in too long chasing thickness and you gain nothing but wasted time and bath drag-out.

⚙ **TECHNICAL STUFF — WHY ALKALINE STRIKES DON’T IMMERSION-PLATE**

In an *acid* copper bath, copper ions are free and eager to swap onto active metal (immersion). In an *alkaline* strike — cyanide or non-cyanide — the copper is chemically **complexed** (chemically “held” by cyanide or another complexer), so it will not spontaneously displace onto the steel. That’s the whole trick: the complexed copper only deposits when you drive it with current, giving a tight, adherent flash instead of loose immersion powder.

• CRITICAL PROCESS CONTROLS

1. COMPLETE COVERAGE BEFORE ACID COPPER

The strike must **fully cover** the part before it moves on. Any bare spot that reaches the acid copper tank will immersion-plate and blister. Watch recesses and low-current areas — they cover last. If the strike doesn't throw into a recess, increase time, improve racking, or check bath balance.

2. LIVE ("HOT") ENTRY

Enter the strike with the **current already on.** If you drop an active part into the strike dead (no current), you get a brief immersion deposit at the instant of insertion — the very thing the strike exists to prevent. Current on, then in.

3. GUARD RINSE AFTER THE STRIKE

A proper rinse follows the strike (often with a mild acid dip to remove any strike-surface oxide) before acid copper. The struck layer plus this rinse together form the pre-plate contamination barrier. **If the strike is cyanide, this rinse is also a life-safety firewall** — cyanide film must be rinsed off before the part goes anywhere near acid.

4. BATH PURITY

Keep the strike free of building-up substrate metal (iron from steel, zinc from die cast). Monitor and treat per TDS; a contaminated strike gives dull, powdery, poorly-adherent flashes.

> ⚠ **HEADS-UP — CYANIDE STRIKE: LETHAL HAZARD (THE MOST IMPORTANT WARNING ON THE LINE)**
> **NEVER allow acid to contact a cyanide bath, cyanide-wet parts, or cyanide-contaminated rinses.** Acid + cyanide releases **hydrogen cyanide (HCN) gas, which can be fatal.** Strict tank segregation, dedicated rinses, and segregated cyanide waste are mandatory. Know where the **cyanide antidote kit** is and who is trained on the emergency protocol before your first shift near this tank `[VERIFY facility protocol]`. The cyanide strike is also strongly alkaline — it causes chemical burns.
>
> OSHA PEL for cyanide (as CN): `[VERIFY]`. If your shop runs the **alkaline non-cyanide** strike, you avoid the HCN hazard entirely — but you still handle a strongly alkaline bath, and you must still know this rule.

> ⚠ **HEADS-UP — WOODS NICKEL STRIKE**
> The Woods strike is **strongly acidic** (concentrated HCl) and contains **nickel** — a skin sensitizer and suspected carcinogen (GHS Category 1A). Full chemical PPE, local exhaust ventilation. OSHA PEL soluble nickel: **1.0 mg/m³ as Ni** `[VERIFY]`. Because it's acidic, keep it and any cyanide chemistry rigorously apart.

> ⚠ **HEADS-UP — RECTIFIER**
> The strike runs on high amperage like any plating tank. **LOTO the rectifier** before any maintenance or reaching into the tank.

STEP-BY-STEP PROCEDURE

1. **Confirm the substrate needs a strike** (steel, zinc die cast, aluminum-after-zincate = yes) and that the part arrived clean, activated, and — for a cyanide strike — free of acid drag-out.
2. **Check the strike bath:** correct type for the job, temperature and pH/CD in range per TDS, anodes present and bagged.
3. **Turn the current ON, then immerse** — live ("hot") entry. Never drop an active part into the strike dead.
4. **Strike for the flash time** (typically 1–5 min) — long enough for *complete coverage*, not for thickness. Verify recesses are covered.
5. **Withdraw and drain** over the tank.
6. **Guard rinse** (and mild acid dip if your process calls for it) before acid copper. For a cyanide strike, rinse thoroughly — no cyanide film may travel toward acid.
7. **Move promptly to acid copper** so the fresh struck surface doesn't oxidize.

TEACH-BACK & TOP MISTAKES — STATION 04

TEACH-BACK — CAN YOU ANSWER?

- Why can't acid copper go straight onto steel? (immersion deposit → blisters)
- Write the immersion reaction ($Cu^{2+} + Fe^0 \rightarrow Cu^0 + Fe^{2+}$).
- Name the two co-equal mainstream strikes (cyanide, alkaline non-cyanide).
- Why enter the strike *live*?
- Why must coverage be *complete* before acid copper?
- State the cyanide rule (acid + cyanide = lethal HCN; never let them meet).
- Roughly how thick is a strike flash? (~2.5–5 μm)

TOP MISTAKES TO AVOID

- Dead entry — a brief immersion deposit forms on insertion.
- Pulling before recesses are fully covered.
- Chasing thickness in the strike (it's a flash).
- Any path that lets acid reach a cyanide tank/rinse.
- Letting iron/zinc build up in the strike bath.
- Skipping the strike on steel/zinc/aluminum "to save a step."

QUICK TROUBLESHOOTING

SYMPTOM	PROBABLE CAUSE	WHAT TO DO
Blistering after acid copper	Immersion deposit; incomplete strike; poor activation	Verify hot entry; extend strike; re-check clean/activation
Strike won't cover recesses	Low CD in recess; short time; low throw	Increase time; improve racking; check bath balance
Dull / powdery strike	Bath out of balance; contamination	Analyze/adjust per TDS; carbon treat if organic
Adhesion failure at strike/basis	Residual oxide or soil under strike	Improve clean + activation; hot entry
Gassy / dark cyanide strike	Low free cyanide; carbonate high	Adjust free cyanide; decarbonate per TDS

SIGN-OFF — STATION 04

Trainee: ______________________ Trainer: ______________________ Strike type: ______________

"I confirm I entered the strike live, achieved complete coverage as a thin flash, guard-rinsed before acid copper, and — for a cyanide strike — kept acid and cyanide rigorously segregated, with required PPE worn."

PART THREE — THE MAIN LINE (STATIONS 05–08)
STATION 05 OF 08 · THE MAIN TANK

ACID COPPER PLATING

WHERE A STRUCK PART BECOMES BRIGHT, LEVELED COPPER

Take a breath. Up to now, every station — cleaning, rinsing, activating, striking — has had one job: get the surface ready. None of those tanks laid down real copper. This tank is the payoff. **Station 05 is where the copper coating is actually created** — the leveling, the brightness, the throw. Everything before it was preparation; everything after it is protection and finishing. If you only deeply understand one station on the whole line, make it this one.

Here's the big idea in one sentence: **we use electricity to pull dissolved copper out of a liquid and stick it, atom by atom, onto your part.** Meet the cast of characters:

- **The cathode** — that's *your part.* **Cathode = the part = negative = where copper lands.**
- **The anode** — phosphorized copper balls in titanium baskets on the positive side. As we plate, they dissolve and feed fresh copper back into the liquid.
- **The electrolyte** — the bath: water loaded with copper sulfate, sulfuric acid, a trace of chloride, and the additive package. The highway the copper travels on.

☀ **FUN FACT**

Because the anodes dissolve to replace the copper that plates out, an acid copper bath is almost self-feeding for metal — as long as the anodes keep their phosphorus film. You spend most of your maintenance effort on acid, chloride, and additives, not on the copper itself.

WHY ACID COPPER? (AND WHY IT'S DEMANDING)

ADVANTAGE	WHAT IT MEANS ON THE FLOOR
Best-in-class leveling	Fills scratches and polishing lines — the reason it undercoats decorative Ni-Cr
Excellent throwing power (high-acid)	Plates uniformly into recesses, through-holes, and vias — the backbone of PCB copper
High cathode efficiency (~95–100%)	Almost every amp you pay for deposits copper; minimal hydrogen
Ductile, low-stress deposit	Soft copper — good for forming and for PCB thermal-cycle reliability
Ambient operation, cheap main bath	Runs at room temperature; simple, non-cyanide metal chemistry

★ **REMEMBER**

The one-sentence identity of this tank: *“Simple metal chemistry. Ruthless on chloride, particulate, and anode film. Control those three and the copper is flawless.”*

That same excellence comes with a demanding personality. Acid copper **will not plate active metals** (that's why you struck the part); it is **extremely sensitive to particulate** (the #1 roughness cause); its **chloride window is narrow** (50–80 ppm); its **organic additives break down** and need carbon treatment; and it needs **phosphorized anodes with a healthy film.** Master those and you've mastered the tank.

⚠ **HEADS-UP (SAFETY & EQUIPMENT)**

The bath is **sulfuric-acid strong** and toxic to aquatic life. Use PP/PVC/lined tanks — sulfuric acid attacks mild steel. Air agitation makes acid/copper-sulfate mist — local exhaust and mist control required. The rectifier is high-amperage — LOTO before maintenance. Always add acid to water.

• THE BATH RECIPE: WHAT'S IN THE TANK AND WHY

The acid copper bath is a balanced recipe of four kinds of ingredients. Know what each one *does*, and you'll know what goes wrong when it drifts.

1. COPPER SULFATE (THE METAL THAT DOES THE COATING)

Copper dissolved in the bath as **copper sulfate ($CuSO_4 \cdot 5H_2O$). Decorative:** 180–250 g/L · **PCB/high-throw:** lower (Cu 60–120 g/L). It *is* the coating. Too low: burning and powder at high-current areas. Too high: sluggish, and it throws worse.

2. SULFURIC ACID (THE CONDUCTOR AND THROW-MAKER)

Sulfuric acid (H_2SO_4). Decorative: 45–75 g/L · **PCB/high-throw:** 180–240 g/L. It carries current, drives throwing power, and helps the anodes corrode. Remember the master lever from Chapter 2: **high copper / low acid = decorative; low copper / high acid = throw (PCB).**

3. CHLORIDE (THE TRACE THAT RUNS EVERYTHING)

Chloride ion (Cl^-) at just **50–80 ppm** (40–60 ppm for PCB). It partners with the suppressor to refine the deposit and keeps the copper-anode phosphorus film healthy. **Titrate it with silver nitrate** — ppm-level, but the single most sensitive control in the bath.

> ★ **REMEMBER — THE RULE CARD: 50-80 PPM CHLORIDE**
> Below ~40 ppm: anodes polarize, deposit goes dull and coarse, poor throw. Above ~100 ppm: anodes passivate (green/gray film), shed copper powder, roughness. Titrate it; don't guess it. It moves constantly — drag-out loses it, DI make-up dilutes it, HCl-based activation drag-in adds it — so correct proactively.

4. THE ADDITIVE SYSTEM (THE LEVELING & BRIGHTNESS CHEMISTRY)

Three organic additives — **suppressor, accelerator, leveler** — dosed *per your chemistry supplier's TDS* and controlled by CVS and Hull Cell. Full section on the next page.

> 💡 **TIP**
> A memory hook for a decorative bath: **copper sulfate 180–250, sulfuric 45–75, chloride 50–80 ppm, temp ~75–80°F.** Keep those in your head and you can sanity-check any analysis sheet at a glance — and remember the numbers *flip* toward low-copper/high-acid for PCB throw.

> ⚙ **TECHNICAL STUFF — TEMPERATURE**
> Acid copper is an **ambient** bath, target ~75–80°F (range 70–90°F). Unlike the hot cleaner, it usually needs *cooling*, not heating — rectifier and agitation add heat, and in summer a chiller keeps it in range. Too cold: dull, sluggish. Too warm: faster additive breakdown and harder brightness control.

• THE ADDITIVE TRIAD — THREE PLAYERS, ONE TEAM

There are **three additive components**, and each controls a different part of the deposit. They are dosed *per TDS* and replenished based on how much you've plated (amp-hours) and on analysis — not on a clock.

COMPONENT	PLAIN-LANGUAGE JOB	IF LOW	IF TOO MUCH
Carrier / Suppressor (needs chloride)	Polarizes the surface, refines grain, gives luster and macro-throw	Coarse, dull, columnar; poor throw	Reduced brightness; haze; over-polarization
Brightener / Accelerator (e.g. SPS)	Depolarizes high-current areas — brightness, throw, via-fill	Dull, hazy; poor throw; slow fill	Brittle/stressed deposit; over-plates recesses
Leveler (N-bearing)	Smooths the surface, controls low-CD, prevents dog-boning	Poor leveling; scratches telegraph; nodules	Dull/pink edges; reduced throw; thin low-CD

★ **REMEMBER**

Suppressor and chloride are a pair — neither works without the other. Accelerator *builds up* unless it's consumed or dragged out; leveler is *consumed* at the surface. More of any one is not better — every component has a "too high" failure mode as real as its "too low" one.

⚙ **TECHNICAL STUFF — HOW THE PROS MEASURE ADDITIVES: CVS**

Acid copper additives are best controlled by **CVS (Cyclic Voltammetric Stripping)** — the industry-standard quantitative lab method — supplemented by the Hull Cell. Track additions against **amp-hours** (amps × hours of current = how much you've plated) and CVS results, and **carbon-treat periodically** to remove breakdown products before they cause dullness, pitting, and steps. All dosing per supplier TDS.

• YOUR BEST FRIEND: THE HULL CELL

Before you ever dump chemistry into a production tank on a hunch, you test your theory in a tiny wedge-shaped tank first — the **Hull Cell.** You plate one small panel under standard conditions — commonly **267 mL, 2 amps, ~5–10 minutes, ambient, air-agitated** (follow your TDS). Because the cell is wedge-shaped, the panel sees *high current at one end and low current at the other all at once* — a single panel shows the whole current-density range, like an X-ray of your bath. Read it against your supplier's reference panels.

PANEL PATTERN	LIKELY CAUSE
Bright and leveled across the full panel	Bath in balance — you're good
Dull / hazy overall	Low accelerator; organic contamination; low chloride
Rough / nodular (sandpaper feel)	Particulate; anode powder (high chloride / filmless anode)
Burnt / powdery at high CD	Low copper; low agitation; chloride imbalance; brightener low
Coarse / columnar, steps	Chloride too low; suppressor not functioning
Poor leveling (scratches telegraph)	Leveler low; chloride imbalance
Dark / red streaks at low CD	Organic or metallic contamination
Pitting	Organic contamination; low wetter; trapped air

💡 **TIP**

Run a Hull Cell *before* and *after* any chemistry addition. Before, to diagnose. After, to confirm the fix worked — on a 267 mL panel instead of risking your whole production tank. It's the cheapest insurance on the line.

• THE ANODES HAVE A SECRET: THE PHOSPHORUS FILM

Acid copper uses **phosphorized copper anodes** (**0.02–0.08% P**). As they work they grow a thin **black film** (containing Cu_3P) that makes them dissolve smoothly and suppresses cuprous (Cu^+) generation and copper powder. That film is essential, and chloride helps maintain it.

ANODE STATE	CAUSE	RESULT ON YOUR PARTS
Healthy black film	Right phosphorus; chloride in range; conditioned	Smooth, uniform dissolution — good
Filmless	Freshly loaded / unconditioned; chloride too low	Sheds copper particles → **roughness**
Passivated	Chloride too high; current too high; too little anode area	Stops dissolving → copper depletes, voltage climbs → starvation

Anodes go into **titanium baskets**, **bagged** in leached polypropylene so fines can't reach the cathode. Condition new anodes, keep adequate anode area (~2:1 to 1:1 anode:cathode), and hold chloride in range. (Some PCB via-fill systems use **inert anodes**—MMO or Pt-Ti—with external copper dosing instead.)

• CONTAMINATION: THE WATCH LIST

Roughness and dullness usually trace to just a few culprits. Know them and where they sneak in.

ITEM	ACTION LEVEL	EFFECT	CORRECTIVE ACTION
Chloride low	`< 40 ppm`	Dull, coarse; anode polarization	Add HCl/NaCl to target; re-titrate
Chloride high	`> 100 ppm`	Roughness; anode passivation; copper powder	Dilute, or silver treatment (precipitate AgCl), per practice
Particulate	`Any`	Roughness, nodules — the #1 cause	Improve filtration; check anode bags; cover tank
Organic breakdown	`CVS / Hull`	Dullness, brittleness, pitting, steps	Carbon treat; replenish additives
Iron (Fe)	`Monitor`	Reduced efficiency, dullness at high levels	Partial dilution; control strike/steel drag-in
Chromium (Cr^{6+}) trace	`Trace`	Dull/dark deposits, poor coverage	Find the chrome drag-back; treat / dilute

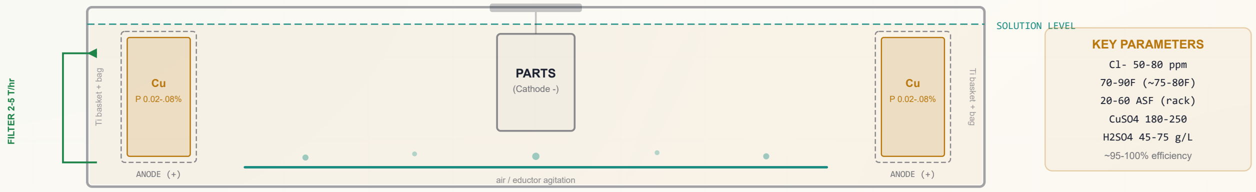
★ **REMEMBER**

Particulate is the #1 cause of roughness in acid copper — which is why the bath runs **continuous filtration, 1–5 µm, at 2–5 turnovers per hour.** A "turnover" is one full tank-volume through the filter. When a part comes out feeling like sandpaper, your first two suspects are *filtration* and *anode film* (via chloride).

⚙ **TECHNICAL STUFF — CHROMIUM DRAG-BACK**

Hexavalent chromium that drifts *backward* from a downstream chrome tank into the copper bath causes dull, dark deposits and poor coverage. The standard removal is a **reduce-then-precipitate** step under your chemist's direction (reduce Cr^{6+} to Cr^{3+}, raise pH to precipitate, filter, re-adjust). Prevention — good rinsing and tank layout — beats cure.

ACID COPPER TANK CROSS-SECTION — TANK ANATOMY



THE SUPPORTING EQUIPMENT (AND WHY EACH PIECE EXISTS)

EQUIPMENT	SETTING / SPEC	WHY IT'S THERE
Anodes	`Phosphorized Cu 0.02–0.08% P, Ti baskets`	Self-replenishing copper; the black film keeps dissolution smooth
Anode bags	`Polypropylene, leached before use`	Catch fines so they don't roughen the deposit
Filtration	`Continuous, 1–5 µm, 2–5 turnovers/hr`	Particulate is the #1 roughness cause — filter it out constantly
Agitation	`Air + cathode rod (rack); eductor for PCB`	Fresh chemistry at the surface; via-fill favors solution flow
Temperature	`70–90°F, target ~75–80°F`	Ambient; often needs cooling, not heating
Current density	`20–60 ASF rack; 5–15 barrel; 15–40 PCB`	Higher CD needs stronger agitation or it burns

STANDARD OPERATING PROCEDURE (STATION 05)

1. **Confirm the part is struck and rinsed** — it should arrive from the strike (Station 04) with complete coverage and no bare active metal. If in doubt, send it back.
2. **Check bath readiness:** chloride 50–80 ppm (titrated); temperature ~75–80°F; agitation and filtration running; anodes present, bagged, filmed, adequate area.
3. **Verify chemistry is current** — copper, acid, chloride, and additives analyzed (CVS/Hull) and replenished per TDS / amp-hours.
4. **Set the rectifier** to the correct current for the part's area and target current density.
5. **Load and plate** — typical time runs from a few minutes to ~60 minutes depending on required thickness.
6. **Monitor while running** — watch for roughness, dullness, unusual voltage climb (anode passivation); confirm agitation and filtration stay on.
7. **At cycle end, drain over the tank** and transfer promptly to the post-plate rinse (Station 06).
8. **Log it** — amp-hours, additions, temperature, chloride. This log is how you dose additives correctly tomorrow.

COMMON DEFECTS — WHAT YOU'LL SEE AND WHAT IT MEANS

WHAT YOU SEE	MOST LIKELY CAUSE	WHAT TO DO
Roughness / nodules (sandpaper)	Particulate; anode powder; high chloride	Improve filtration; check anode bags & film; correct chloride
Burning at high CD	Low copper; low agitation; high CD; chloride imbalance	Add copper; increase agitation; reduce CD; correct chloride
Dull / hazy deposit	Accelerator low; organics; chloride low	Add accelerator; carbon treat; check chloride
Poor leveling	Leveler depleted; chloride imbalance	Add leveler; correct chloride
Pitting	Organic contamination; trapped air; low wetter	Carbon treat; improve agitation/positioning; add wetter
Brittle / cracked deposit	Excess accelerator/leveler; organic breakdown	Rebalance additives; carbon treat
Poor throwing power	Copper too high / acid too low; low additives	Shift Cu:acid toward high-acid regime; check additives
Voltage rising, copper dropping	Anode passivation (high Cl^- / low area / high current)	Correct chloride; add anode area; reduce current
Blistering / adhesion failure	Immersion deposit — strike skipped/incomplete	Fix the strike stage; hot entry; verify activation

SAFETY — THIS STATION DEMANDS YOUR RESPECT

⚠ **HEADS-UP — ACIDIC COPPER SULFATE BATH & MIST**

Sulfuric acid causes **severe burns**; copper sulfate is an irritant and **highly toxic to aquatic life** (waste-treat, never to drain). Air agitation throws an **acid / copper-sulfate mist** — local exhaust ventilation and mist eliminators required. Full PPE: goggles *and* face shield, chemical (nitrile) gloves, apron, boots; respirator if ventilation is inadequate. Flush 15 minutes on contact. OSHA PEL sulfuric acid mist 1 mg/m³ `[VERIFY]`; copper dust/mist 1, fume 0.1 mg/m³ as Cu `[VERIFY]`.

⚠ **HEADS-UP — RECTIFIER & RIGHT TANK, RIGHT METAL**

The rectifier is **high amperage** — **LOTO** before any maintenance, reaching in, or bus-bar work. And **sulfuric acid attacks mild steel** — tanks, heaters/coolers, and fittings are PP/PVC or lined for a reason. Never drop steel tools into the bath: they corrode, dump iron, and can short across electrodes.

TEACH-BACK & TOP MISTAKES — STATION 05

TEACH-BACK — CAN YOU ANSWER?

- Which electrode is your part — and which charge?
- The chloride window, and what low vs. high does?
- The three additives and what each controls?
- The Cu:acid lever — decorative vs. PCB throw?
- What the anode phosphorus film does; filmless vs. passivated?
- The #1 cause of roughness (particulate) and the fix?
- How additives are measured (CVS + Hull Cell)?
- Why blistering points back to the strike?

TOP MISTAKES TO AVOID

- Guessing chloride instead of titrating it.
- Chasing brightness by dumping in more brightener.
- Neglecting filtration (roughness & nodules).
- Running filmless or passivated anodes.
- Ignoring rising voltage (anode passivation).
- Plating a part with incomplete strike coverage.

SIGN-OFF — STATION 05

I confirm that I can independently: verify bath readiness (chloride, temp, agitation, filtration, anodes); read an analysis sheet against target ranges; run and interpret a Hull Cell panel; identify which additive or condition is responsible for a given defect; maintain the anode phosphorus film; and state the copper-sulfate/sulfuric-acid safety precautions. I understand that additive and chemistry adjustments are made only by authorized personnel.

Operator: ____________________ Date: __________ Trainer: ____________________ Date: __________

STATION 06 OF 08

POST-PLATE RINSE & DRAG-OUT RECOVERY

RECOVER THE COPPER — IT'S MONEY GOING OUT, AND A CONTAMINANT GOING FORWARD

Your part just came out of the acid copper bath dripping with copper and sulfuric acid. That film is two things at once: **copper you paid for** (money), and a **contaminant** to whatever comes next. As the saying goes: *"Copper sulfate dragged into the nickel tank is both a dollar lost and a defect waiting to happen."* This station recovers the copper and protects the downstream chemistry.

> ★ **REMEMBER**
> A rinse is never "just a rinse." Each one is a *checkpoint* that protects the next tank. This one does double duty: it **recovers copper drag-out** for return to the bath, and it **keeps copper and acid out of the downstream nickel or post-treatment.**

WHAT AND WHY

Acid copper drag-out carries copper and sulfuric acid into the next process. Into a **nickel bath, copper is a poison at just a few ppm** — it makes dark, pitted low-current areas. So we use a smart two-part approach: a **drag-out recovery rinse** (often a still first rinse) captures the copper-rich film so it can be periodically returned to the plating bath, followed by a **clean rinse** that polishes the part down to a low conductivity before it moves on.

OPERATOR ESSENTIALS

PARAMETER	RANGE	NOTES
Temperature	`Ambient`	No heating required
Time	`30–60 sec with agitation`	Double rinse recommended
Conductivity	`< 200 µS/cm (target < 100)`	Copper/acid drag-out is the concern
Drain time	`15–20 sec over the copper tank`	Maximizes drag-back recovery
Drag-out recovery	`Optional still first rinse`	Periodically return to the copper bath
Water	`DI preferred`	Avoids hard-water spotting on fresh copper

> 💡 **TIP — THE MAGIC OF THE DOUBLE COUNTER-CURRENT RINSE**
> Instead of one rinse tank, use two in series. Parts move *forward* (dirty → clean), while fresh water flows *backward* (clean → dirty) over a weir. Parts get their final rinse in the cleanest water, while you use far less water overall — and you can put a still recovery rinse first to catch the copper.

```
FROM        +-------------+  PARTS ->  +-------------+        TO
COPPER ->   |  RECOVERY   | ---------> |  RINSE 2    | ---->  NEXT
            |  (still)    |            |  (clean)    |       STAGE
            | Cu-rich ->  |            | <200 uS/cm  |
            | back to Cu  |            +-------------+
            +-------------+                  ^
                 ^   overflow <----------  |  <- FRESH IN
```

> ⚠ **HEADS-UP — SAFETY**
> Dilute acidic copper solution — an irritant and toxic to aquatic life (waste segregation). Gloves at all times when handling parts/racks; non-slip footwear on the wet floor.

FRESH COPPER TARNISHES FAST

★ **REMEMBER**

Freshly plated bright copper **oxidizes and tarnishes rapidly in air.** Keep parts *wet* and move quickly to the next stage. Water spots and oxidation on fresh copper telegraph into the subsequent nickel — or become visible on a bare-copper finish. If parts must wait, an **anti-tarnish (benzotriazole-based) dip** preserves the finish.

STANDARD OPERATING PROCEDURE

1. **Drain over the copper tank** a full 15–20 seconds first — this is where you recover the most copper.
2. **Dip the recovery (still) rinse** if equipped, then the clean rinse, moving dirty → clean.
3. **Immerse with agitation, 30–60 seconds,** flushing blind holes and recesses (trapped pockets hold copper and acid).
4. **Confirm conductivity is < 200 µS/cm** before the part moves on.
5. **Handle with clean gloves only** — bare fingers etch permanently into fresh copper.
6. **Move promptly** to post-treatment (Station 07) before the copper oxidizes; keep it wet.

QUICK TROUBLESHOOTING

SYMPTOM	PROBABLE CAUSE	WHAT TO DO
Copper climbing in the nickel bath	Copper drag-in	Tighten rinse; add recovery + second rinse; longer drain
Water spots / haze on copper	Hard water; slow transfer; oxidation	Use DI; keep wet; speed transfer
Stains on fresh copper	Air oxidation during a delay	Anti-tarnish dip; reduce transfer time
Roughness carried forward	Particulate not rinsed off	Improve rinse agitation; check upstream filtration

TEACH-BACK CHECKLIST

- [ ] Explain the two reasons this rinse matters (recover copper = money; keep copper out of downstream nickel = quality).
- [ ] State why copper is a problem in a nickel bath (poisons low-current areas at a few ppm).
- [ ] State the conductivity target (< 200 µS/cm, aim < 100) and the drain time for recovery (15–20 s over the copper tank).
- [ ] Explain why fresh copper is kept wet and handled with clean gloves (it tarnishes and etches fast).
- [ ] Describe a drag-out recovery rinse and how the copper gets back to the bath.

SIGN-OFF — STATION 06

Trainee: ____________________ Trainer: ____________________ Conductivity: __________ µS/cm

“I confirm the post-plate rinse conductivity was within limit (< 200 µS/cm), copper drag-out was drained/recovered, parts were kept wet and handled with clean gloves, and PPE was worn.”

STATION 07 OF 08 (+ STAGE 8 FINAL RINSE/DRY)

POST-TREATMENT / QC & THE BAKE DECISION

COPPER RARELY STANDS ALONE — DECIDE WHAT IT BECOMES, THEN VERIFY IT

You've made bright, leveled copper — but copper is rarely the finished product. As the tagline says: *"Bare copper is beautiful for about a day. What comes next decides whether it lasts."* This station is where you decide the copper's role, apply any protection or handoff, and — on hardened steel — make the hydrogen-embrittlement bake decision.

★ **REMEMBER — THE THREE ROLES COPPER PLAYS NEXT**
Acid copper is usually one of three things: an **undercoat** (on to bright nickel + chrome in the decorative triad), a **functional / PCB deposit** (verified and finished), or a **decorative bare-copper / antique finish** (which must be protected, because copper tarnishes fast). Know which role applies *before* you leave this stage.

POST-TREATMENT / HANDOFF OPTIONS

PATH	TYPICAL USE	KEY ACTION
Undercoat → Bright Nickel + Chrome	Decorative triad (auto trim, plumbing, hardware)	Activate copper, proceed to nickel; copper's leveling carries the system
Anti-tarnish (benzotriazole / chromate)	Bare copper stored or shipped	Immersion dip per TDS; prevents oxidation
Antique / oxidized copper finish	Decorative art, lighting, hardware	Sulfide/selenide blackening + relieving + lacquer
Clear lacquer / organic topcoat	Bright bare-copper decorative	Spray/dip; air or heat cure
PCB — no coating (or ENIG/OSP later)	Circuit fabrication	Move to imaging/etch or final finish per process

💡 **TIP — THE NICKEL HANDOFF IS TIME-SENSITIVE**
If copper is going straight to nickel, **keep it wet, move fast, and re-activate** the copper before nickel. Copper that oxidizes before the nickel goes on causes nickel haze and, at worst, nickel that peels. Most "nickel-over-copper" defects are really *copper that sat too long*.

⚠ **HEADS-UP — SAFETY DEPENDS ON THE PATH**
Anti-tarnish / blackening chemistries vary — some contain **selenium or sulfide** compounds; consult the SDS. If a **hexavalent chromate** anti-tarnish is used, Cr^{6+} is a confirmed carcinogen (OSHA PEL 5 µg/m³ `[VERIFY]`) — full respiratory protection and controls apply. Lacquer solvents are flammable — ventilation required. Standard chemical PPE throughout.

QUALITY CONTROL CHECKS

CHECK	METHOD	REFERENCE
Thickness	XRF; microsection	`ASTM B568; ASTM B487`
Adhesion	Bend, thermal shock, peel	`ASTM B571`
Leveling / brightness	Visual + profilometer; Hull Cell	`Internal / customer spec`
Ductility / elongation (PCB)	Foil tensile / elongation	`IPC-TM-650`
Solderability (PCB/electronics)	Wetting balance / dip	`J-STD-002 [VERIFY rev]`
Porosity (bare copper)	Corrosion / porosity test	`Per spec`

TECHNICAL STUFF — THE SPECS MAY HAVE MOVED

Engineering copper and PCB work reference standards such as **ASTM B734** (engineering copper), **ASTM B456** (decorative Cu-Ni-Cr), **AMS 2418** / legacy **MIL-C-14550** (copper plating), and **IPC-6012 / IPC-A-600** (PCB acceptability). Standards get revised — always confirm the current issue against the customer callout. `[VERIFY current revisions of AMS 2418 / MIL-C-14550, IPC-6012, J-STD-002, ASTM B850.]`

THE HYDROGEN-EMBRITTLEMENT BAKE (CRITICAL FOR HARD STEEL)

Some jobs require baking, and this is the natural decision point. High-strength steel absorbs atomic hydrogen during pickling and plating, and that hydrogen can cause delayed cracking. **Baking** drives it back out before it can do damage.

HEADS-UP (SAFETY-CRITICAL — THIS PREVENTS CATASTROPHIC PART FAILURE)

For hardened steel **above 40 HRC**: bake **within 4 hours** of the last electrodeposition step, at **375 ± 25°F**, for a **minimum of 8 hours**, per **ASTM B850** `[VERIFY rev]`. The 4-hour clock starts at completion of the last plating step. Check the customer spec for whether the bake occurs *before* subsequent nickel/chrome. A missed bake on a structural fastener isn't a cosmetic defect — it's a part that can snap in service. Never skip it on embrittlement-sensitive work. (Testing per ASTM F519 where required.)

TROUBLESHOOTING (STATION 07)

SYMPTOM	PROBABLE CAUSE	WHAT TO DO
Copper tarnishes / darkens	Expected — copper oxidizes fast	Apply anti-tarnish / lacquer; reduce exposure
Nickel haze over copper	Copper oxidized before nickel; slow transfer	Activate copper; speed transfer; keep wet
Nickel peels from copper	Passive/oxidized copper surface	Acid-activate copper before nickel
Fingerprints etched in copper	Bare-handed handling of fresh copper	Clean gloves only
Poor solderability (PCB)	Oxidation; organic residue; wrong finish	Verify finish/OSP; clean; re-test

TEACH-BACK CHECKLIST

- [ ] Name the three roles copper plays after plating (undercoat, functional/PCB, protected bare-copper).
- [ ] State why a bare-copper finish needs anti-tarnish or lacquer.
- [ ] State the HE bake parameters exactly (375 ±25°F, ≥8 h, within 4 h of the last plating step, ASTM B850).
- [ ] Explain why copper going to nickel must be kept wet and re-activated.
- [ ] Name two QC checks and their references (e.g., thickness by ASTM B568/B487; adhesion by ASTM B571).

SIGN-OFF — STATION 07

I confirm that I can independently: identify the copper's role and select the correct handoff / post-treatment; apply the correct PPE for the chosen path; state and apply the ASTM B850 bake requirement for hardened steel (375 ±25°F, ≥8 h, within 4 h); run the required QC checks against spec; and keep fresh copper protected and clean-handled.

Operator: ____________________ Date: __________ Trainer: ____________________ Date: __________

STAGE 08 OF 08 · THE FINISH LINE

FINAL RINSE & DRY

THE LAST 60 SECONDS — WHERE A PERFECT PART IS FINISHED, OR QUIETLY SPOTTED AND STAINED

The copper is on and the post-treatment decided. You are one short stage from a finished part. Stage 08 has two simple jobs: **rinse off the leftover chemistry** so it doesn't leave residue, and **dry the part gently** without water spots or oxidation. It feels like the easy part — and that is exactly why parts get wrecked here. Fresh copper is reactive; a slow, hot, or bare-handed finish undoes everything the line just accomplished.

★ **REMEMBER**
Fresh copper **oxidizes and water-spots fast, and fingerprints etch permanently.** Speed, DI water, clean gloves, and gentle heat protect every step upstream. If the finish is bare copper, its protection (anti-tarnish or topcoat) should already be on from Station 07 — don't leave it bare.

OPERATOR ESSENTIALS

STEP	SETTING	WHY IT'S SET THERE
Final rinse	`DI water, ambient, 15–30 sec, gentle immersion`	Removes residue without spotting the fresh copper
Dry temperature	`120–140°F max, warm air`	Drives off water fast before it can spot or oxidize
Handling	`Clean, dry, lint-free gloves only`	Bare fingers etch permanently into fresh copper
Protection	`Anti-tarnish / topcoat if bare copper`	Copper tarnishes within a day if left bare

⚠ **HEADS-UP (DON'T RUIN A GOOD PART AT THE FINISH LINE)**
Two easy ways to spoil a perfect copper deposit in the last 60 seconds: **letting it air-dry with hard water on it** (water spots and oxidation stains that telegraph into nickel or show on bare copper), and **handling it bare-handed** (etched fingerprints you can't polish out). Warm-dry promptly, use DI, and wear clean gloves.

STANDARD OPERATING PROCEDURE (STAGE 08)

1. **Final DI rinse:** gentle immersion in ambient DI water for **15–30 seconds.**
2. **Drain** over the rinse tank so you carry off as little water as possible.
3. **Dry gently** in warm air, **120–140°F max** — fast enough to beat spotting, gentle enough not to discolor.
4. **Confirm protection** is in place if the finish is bare copper (anti-tarnish/lacquer from Station 07).
5. **Handle with clean gloves** throughout, and inspect for spotting, haze, or tarnish; verify the finish per spec.

TEACH-BACK CHECKLIST

A trainee is signed off on Stage 08 only when they can, unprompted:

- [ ] State the two jobs of Stage 08 (rinse off residue; gently dry without spotting).
- [ ] State the final-rinse water, temperature, and time (DI, ambient, 15–30 sec).
- [ ] State the maximum dry temperature and why warm-air drying is done promptly (120–140°F; beat water spots and oxidation).
- [ ] Explain why fresh copper is handled with clean gloves only (fingerprints etch permanently).
- [ ] State how a bare-copper finish is protected (anti-tarnish or topcoat).

SIGN-OFF — STAGE 08

Trainee: ______________________ Trainer: ______________________ Date: ____________

"I confirm the part received a final DI rinse, was warm-dried at 140°F or below without water spots, was handled with clean gloves, and (if bare copper) was protected against tarnish — with required PPE worn."

PART FOUR — BACK MATTER & CERTIFICATION

YOUR DECODER RING

GLOSSARY — EVERY TERM, PLAIN AND SIMPLE

ANYTIME A WORD IN THIS MANUAL MADE YOU PAUSE, FLIP BACK HERE

> **TIP**
> Keep this glossary tabbed. The fastest way to sound like a veteran on the shop floor is to use the right word for the right thing.

Accelerator (Brightener): The additive that speeds deposition in high-current areas — brightness, throw, and via-fill. Often SPS. Builds up unless consumed/dragged out; too much embrittles.

Acid Activation (acid dip, pickle): Station 3. A dip in acid (H_2SO_4 or HCl) that dissolves surface oxide and "wakes up" bare metal so the strike/copper can bond. *No activation, no adhesion.*

Acid Copper Sulfate: The process this manual is about. Copper sulfate + sulfuric acid + trace chloride + organic additives. Best leveling and throw of the common baths, but won't plate active metals directly.

Adhesion: How well the copper sticks to the substrate. Good adhesion means it won't blister or peel.

Additive System: The organic package that levels and brightens acid copper: **suppressor, accelerator, leveler.** Controlled by CVS and Hull Cell.

Agitation: Movement of solution or parts. Keeps fresh chemistry against the part and evens the deposit. Air, cathode-rod, eductor (PCB), or barrel rotation.

Alkaline: The opposite of acidic; high pH (above 7). The cleaner and the copper strikes are alkaline; the acid copper bath is not.

Alkaline Non-Cyanide Copper Strike: A cyanide-free strike (often pyrophosphate/complexed) for steel and zinc. Treated as **co-equal** with the cyanide strike in this manual.

Ambient: Room temperature, roughly 60–85°F. The acid copper bath runs near ambient (~75–80°F).

Amp-Hour (A·h): "How much plating you've done": amps × hours. Additives are consumed per amp-hour, so you replenish based on amp-hours and analysis, not clock time.

Anode: The positively-charged metal in the tank. Acid copper uses **phosphorized copper** anodes that dissolve to feed the bath.

Anode Bag: A leached polypropylene bag around each anode/basket that catches fines so they don't roughen the deposit. *Always bag your anodes.*

ASF: Amps per Square Foot — a unit of current density. Decorative acid copper runs about 20–60 ASF (rack).

ASTM B456: Spec for decorative Cu+Ni+Cr systems — copper as the leveling undercoat of the decorative triad.

ASTM B487 / B568: Coating-thickness methods — microscopical cross-section (B487) and X-ray/XRF (B568). `[VERIFY rev]`

ASTM B571: Qualitative adhesion testing (bend, thermal shock, peel) for copper. `[VERIFY rev]`

ASTM B850: The spec covering the hydrogen-embrittlement relief bake (post-plate oven treatment for hardened steel). `[VERIFY rev]`

AMS 2418 / MIL-C-14550: Copper plating specs (aerospace/defense; legacy military). Verify current callout. `[VERIFY rev]`

Barrel Plating: Plating many small parts by tumbling them in a rotating perforated barrel. Contrast with Rack Plating.

Carbon Treatment: A bath cleanup where you circulate the solution through activated carbon to pull out organic contamination and broken-down additive.

Carrier / Suppressor: The additive that polarizes the surface, refines grain, and provides luster and macro-throw. **Only works with chloride present.**

Cathode: The part being plated — the negatively-charged work. Copper deposits onto the cathode.

Cathode Efficiency: The percentage of current that actually deposits metal. Acid copper is **~95–100%** — nearly all current makes copper.

Chloride (Cl^-): A trace ingredient (**50–80 ppm**) that partners with the suppressor to refine the deposit and maintains the anode phosphorus film. Titrated with silver nitrate. Too low = dull/coarse; too high = anode passivation/powder.

Chromium Drag-Back: Hexavalent chromium drifting backward from a downstream chrome tank into the copper bath — causes dull, dark deposits. Prevent with good rinsing/layout.

Complexing (Chelating): Chemically "holding" metal ions in solution. In alkaline strikes, copper is complexed so it won't immersion-deposit on steel.

Conductivity (µS/cm): A measure of dissolved salts in rinse water — how "dirty" the rinse is. **Lower is cleaner.**

Counter-Current Rinse: A water-saving setup where fresh water flows opposite to the parts, so the cleanest water meets the cleanest parts.

CVS (Cyclic Voltammetric Stripping): The industry-standard quantitative lab method for measuring acid copper additives, supplemented by the Hull Cell.

Cyanide Copper Strike: A strongly alkaline copper strike with excellent adhesion. **Cyanide is lethal; acid contact releases HCN gas.** Co-equal option with the non-cyanide strike.

Decorative Triad: The standard decorative system per ASTM B456: **Copper (leveling) → Bright Nickel → Chromium.**

Dezincification: Selective removal of zinc from brass by over-aggressive acid, leaving a weak coppery smut. Keep brass dips short (5–15 s).

DI Water (Deionized Water): Water with dissolved minerals removed. Cleaner than tap; preferred near the plating tank and for PCB work.

Drag-In / Drag-Out: Chemistry that rides along on the parts from one tank to the next. *Drag-out* leaves a tank; *drag-in* arrives in the next. Rinses control it.

Dummy / Dummying: A cleanup trick: plate onto scrap "dummy" cathodes at low current to draw contamination out of the bath.

Electroclean: Cleaning with current through the part to generate scrubbing gas. Finish **anodic** to avoid metallic smut.

Filtration: Continuous pumping of the bath through a filter (1–5 µm, 2–5 turnovers/hr). **Particulate is the #1 roughness cause** in acid copper.

• GLOSSARY (CONTINUED)

Hull Cell: A small, wedge-shaped test tank that plates a sample panel across a *range* of current densities at once. Reading the panel tells you what's wrong with the bath. Standard test: **267 mL, 2 A, ~5–10 min, ambient, air-agitated.**

Hydrogen Embrittlement (HE): When atomic hydrogen soaks into steel during pickling/plating and makes hardened steel **brittle** — it can crack later under load, sometimes catastrophically.

HE Relief Bake: Baking plated parts to drive hydrogen back out. For steel **>40 HRC**: bake **within 4 hours** of the last plating step at **375 ±25°F for at least 8 hours** (per ASTM B850 `[VERIFY rev]`).

HRC (Rockwell C Hardness): A hardness scale for steel. Above **40 HRC**, steel is "high-strength" and at real risk of hydrogen embrittlement.

Immersion Deposit (Galvanic Displacement): Loose, non-adherent copper that forms *chemically* when an active metal (steel, zinc, aluminum) contacts acid copper — before any current. The reason a **strike** is required. $Cu^{2+} + Fe^0 \rightarrow Cu^0 + Fe^{2+}$.

Inhibitor: An additive in the acid pickle that slows attack on the base metal and reduces hydrogen absorption. **Required for >40 HRC steel.**

IPC-6012 / IPC-A-600: PCB qualification and acceptability standards. **IPC-TM-650:** PCB test methods (copper elongation/tensile, throw). `[VERIFY rev]`

Iron (Fe) Contamination: Dissolved iron dragged into the bath from steel or the strike. Reduces efficiency and dulls the deposit at high levels. Control strike/steel drag-in.

J-STD-002: Solderability test standard for PCB/electronics copper. `[VERIFY rev]`

Leveler: The additive that smooths the surface and controls low-current areas (prevents "dog-boning"). Nitrogen-bearing; consumed at the surface.

Leveling: The bath's ability to fill scratches and produce a surface *smoother* than the substrate. Acid copper levels better than any common bath — its signature strength.

LOTO (Lockout/Tagout): De-energizing and locking the power off before working on the rectifier or electrical equipment.

µm (Micrometer / Micron): A unit of thickness. One micron = 0.001 mm ≈ 0.04 mil. A copper strike is **~2.5–5 µm**; a plated copper build is thicker.

Mil: One-thousandth of an inch (0.001″). 1 mil ≈ 25 µm.

Oxide: The thin tarnish/film that forms on bare metal in air. It blocks adhesion, so acid activation must dissolve it right before the strike.

PCB (Printed Circuit Board): Electronics substrate where high-throw acid copper plates uniformly through holes and fills vias. Runs low-copper / high-acid.

pH: A 0–14 scale of acidity. Below 7 = acidic. The acid copper bath is strongly acidic from sulfuric acid; the strikes are strongly alkaline.

Phosphorized Copper Anode: Copper alloyed with 0.02–0.08% phosphorus. Grows a protective black film (Cu_3P) for smooth dissolution. Filmless → powder; passivated → starvation.

Phosphorus Film (Cu_3P): The thin black film on a healthy phosphorized copper anode that keeps dissolution smooth and suppresses copper powder.

Pitting: Tiny holes in the deposit, usually from organic contamination, trapped air/gas, or low wetter.

ppm (Parts Per Million): A unit for very small concentrations — used for the chloride level (50–80 ppm).

Rack Plating: Plating parts individually hung on a rack/fixture. Used for larger or cosmetic parts — most decorative copper.

Rectifier: The power supply that converts AC into the DC current that drives plating. **High amperage — shock and arc-flash hazard.** Always LOTO before servicing.

RoHS: "Restriction of Hazardous Substances." The non-cyanide strike and the acid copper bath support RoHS-compliant, cyanide-free finishing.

SPS: Bis-(sodium sulfopropyl)-disulfide — a common accelerator/brightener in acid copper.

Strike: A thin, tightly-bonded flash (copper or nickel) plated under controlled conditions to give acid copper a noble, adherent surface on active metals. Enter *live*; get *complete coverage.*

Sulfuric Acid (H_2SO_4): The acid in the copper bath (and one activation option). Carries current, drives throwing power, corrodes the anodes. Causes severe burns; add acid to water.

Throwing Power: The bath's ability to plate evenly into recesses and through-holes. Acid copper's throw is **excellent** in high-acid formulations — the basis of PCB copper.

Titrate / Titration: A chemical test that measures a solution's concentration. Chloride is titrated with silver nitrate; the cleaner is titrated for strength.

Turnover (Tank Turnover): One complete pass of the entire tank volume through the filter. Acid copper runs 2–5 turnovers/hour.

Water Break Test: The free, reliable test for a clean surface: dip and pull the part — if water **sheets** off unbroken it's clean; if it **beads**, oil remains, so reclean.

Wetting Agent (Wetter): An anti-pitting surfactant, often part of the carrier/additive package; too little wetter → pitting.

Woods Nickel Strike: A strongly acidic nickel-chloride/HCl strike used to activate stainless and difficult steels before copper. Nickel is a skin sensitizer / suspect carcinogen.

FIELD REFERENCE

TROUBLESHOOTING QUICK-INDEX

FIND YOUR SYMPTOM, READ THE LIKELY CAUSE, TRY THE FIX

★ **REMEMBER**
Most acid copper defects trace back to **three root causes**: chloride out of window, particulate/anode-film trouble, or an additive imbalance — plus one adhesion killer, a skipped or incomplete strike. Diagnose with a **Hull Cell** (and CVS) before you start dumping chemistry.

⚠ **HEADS-UP**
Before changing acid strength or pickle time on hard steel, check whether the part is **>40 HRC** — those changes raise hydrogen-embrittlement risk. And never troubleshoot a cyanide strike near acid: **acid + cyanide = lethal HCN.**

STATION 01 — SOAK CLEAN / ELECTROCLEAN

SYMPTOM	LIKELY CAUSE	FIX
Water break fails (water beads)	Low concentration or temp	Titrate cleaner; increase temperature
Oily film remains after cleaning	Floating oil redepositing; exhausted cleaner	Skim/dump; recharge cleaner
Etching/discoloration of zinc die cast	Cleaner too alkaline/hot	Reduce temp; die-cast-safe cleaner
Skip plating / bare spots downstream	Passive film / fingerprints; ended cathodic	Finish electroclean anodic; clean gloves

STATION 02 — RINSE (PRE-ACTIVATION)

SYMPTOM	LIKELY CAUSE	FIX
Rinse conductivity high (>500 µS/cm)	Too much drag-out; low flow	More drain time off cleaner; more rinse flow
Acid activation tank goes alkaline	Cleaner carrying over	Improve rinse; add a second stage
Foaming in rinse	Surfactant from cleaner	Increase water flow; check cleaner

STATION 03 — ACID ACTIVATION

SYMPTOM	LIKELY CAUSE	FIX
Dark oxide still on steel	Acid too dilute; dip too short	Increase conc.; extend time **(check HE risk)**
Pitting / etching of base metal	Too concentrated/long/warm	Reduce conc.; shorten dip; add inhibitor
Copper smut on brass	Dezincification (over-dipped)	Shorten dip; go to a copper strike
Adhesion fails downstream	Incomplete activation; alkaline drag-in	Re-check acid strength/time; tighten Station 02
HE cracking / failures	Over-long pickle; no inhibitor; >40 HRC	Shorten pickle; inhibitor; relief bake

STATION 04 — COPPER STRIKE

SYMPTOM	LIKELY CAUSE	FIX
Blistering after acid copper	Immersion deposit; incomplete strike; poor activation	Verify hot entry; extend strike; re-check clean/activation
Strike won't cover recesses	Low CD in recess; short time; low throw	Increase time; improve racking; check balance
Dull / powdery strike	Bath out of balance; contamination	Analyze/adjust per TDS; carbon treat
Gassy / dark cyanide strike	Low free cyanide; carbonate high	Adjust free cyanide; decarbonate per TDS

STATION 05 — ACID COPPER PLATING

SYMPTOM	LIKELY CAUSE	FIX
Roughness / nodules (sandpaper)	Particulate; anode powder; high chloride	Improve filtration; check anode bags/film; correct chloride
Burning at high CD	Low copper; low agitation; high CD; chloride imbalance	Add copper; increase agitation; reduce CD; correct chloride
Dull / hazy deposit	Accelerator low; organics; chloride low	Add accelerator; carbon treat; check chloride
Poor leveling	Leveler depleted; chloride imbalance	Add leveler; correct chloride
Pitting	Organic contamination; trapped air; low wetter	Carbon treat; improve agitation; add wetter
Brittle / cracked deposit	Excess accelerator/leveler; organic breakdown	Rebalance additives; carbon treat
Poor throwing power	Copper high / acid low; low additives	Shift Cu:acid to high-acid regime; check additives
Voltage rising, copper dropping	Anode passivation (high Cl^- / low area / high current)	Correct chloride; add anode area; reduce current
Blistering / adhesion failure	Immersion deposit — strike skipped/incomplete	Fix the strike; hot entry; verify activation

STATION 06 — RINSE (POST-PLATE)

SYMPTOM	LIKELY CAUSE	FIX
Copper climbing in the nickel bath	Copper drag-in	Tighten rinse; add recovery + second rinse
Water spots / haze on copper	Hard water; slow transfer; oxidation	Use DI; keep wet; speed transfer
Roughness carried forward	Particulate not rinsed off	Improve rinse agitation; check upstream filtration

STATION 07 — POST-TREATMENT / QC

SYMPTOM	LIKELY CAUSE	FIX
Nickel haze / peel over copper	Copper oxidized before nickel; slow transfer	Re-activate copper; keep wet; speed transfer
Copper tarnishes / darkens	Expected — copper oxidizes fast	Anti-tarnish / lacquer; reduce exposure
Poor solderability (PCB)	Oxidation; organic residue; wrong finish	Verify finish/OSP; clean; re-test

HULL CELL QUICK-READ (267 ML · 2 A · ~5–10 MIN · AMBIENT, AIR-AGITATED)

PANEL PATTERN	LIKELY CAUSE
Bright and leveled across the full panel	Bath in balance
Dull / hazy overall	Low accelerator; organics; low chloride
Rough / nodular	Particulate; anode powder (high Cl^- / filmless anode)
Burnt / powdery at high CD	Low copper; low agitation; chloride imbalance
Coarse / columnar, steps	Chloride too low; suppressor not working
Dark / red streaks at low CD	Organic or metallic contamination

💡 **TIP**

Notice how many defects trace back to **chloride** and **anodes**. When in doubt, titrate the chloride and look at your anode film first — they're behind more acid copper problems than anything else.

TEMPLATE — PRINT ONE PER OPERATOR

OPERATOR TRAINING MATRIX

HOW A SUPERVISOR PROVES — ON PAPER — THAT EACH OPERATOR IS TRAINED AND SIGNED OFF BEFORE RUNNING SOLO

★ **REMEMBER**

An operator is "line-qualified" only when every station they run is signed. Nobody touches the **rectifier**, the **acid activation**, the **strike**, or the **copper bath** solo until the **Safety / PPE & SDS** row — and, where a cyanide strike is used, the **cyanide handling** row — are signed. Safety qualification comes first, always.

Sign-off levels: O = Observed only · **A** = Assisted under supervision · **Q** = Qualified to run solo · **T** = Qualified to Train others.

#	STATION / SKILL	O/A/Q/T	TRAINER INIT & DATE	OPERATOR INIT & DATE	RE-CERT DUE
1	Safety / PPE & SDS (all stations)	____	________	________	_____
2	Cyanide handling & acid segregation (if used)	____	________	________	_____
3	Station 01 — Soak Clean / Electroclean + water break	____	________	________	_____
4	Station 02 — Rinse (Pre-Act) + conductivity	____	________	________	_____
5	Station 03 — Acid Activation (incl. HE check)	____	________	________	_____
6	Station 04 — Copper Strike (live entry, coverage)	____	________	________	_____
7	Station 05 — Acid Copper Plating operation	____	________	________	_____
8	Chloride titration + Hull Cell / CVS interpret	____	________	________	_____
9	Anode management (phosphorus film, bagging)	____	________	________	_____
10	Station 06 — Post-Plate Rinse & drag-out recovery	____	________	________	_____
11	Station 07 — Post-Treatment / QC & handoff	____	________	________	_____
12	HE Bake (ASTM B850) decision & logging	____	________	________	_____
13	Stage 08 — Final Rinse / Dry	____	________	________	_____
14	LOTO on rectifier	____	________	________	_____
15	Emergency response — eyewash, shower, spill, cyanide	____	________	________	_____
16	Waste / drag-out handling (Cu toxicity; CN segregation)	____	________	________	_____
17	Troubleshooting — symptom recognition	____	________	________	_____

Operator name: ______________________________ **Hire/start date:** __________________

Supervisor name: ______________________________ **Review cycle:** ☐ Annual ☐ Other: ______________

⚠ **HEADS-UP**

Safety rows (1, 2, 14, 15) are not optional and not "fill in later." An operator may not work *any* station until PPE/SDS, cyanide handling (where used), LOTO, and emergency response are signed off. Safety competency comes before production competency, always.

💡 **TIP**

Re-verify each operator annually, and **immediately** after any process or chemistry change. Note the recertification date in the cell (e.g., "Q 7/26 · recert 7/27"). People drift from procedure over time, and a yearly refresher catches bad habits before they become defects — or injuries.

20 QUESTIONS · PASSING SCORE 80% (16/20)

ACID COPPER COMPLETION TEST

COVERS EVERY STATION PLUS SAFETY — ANSWER IN YOUR OWN WORDS WHERE ASKED

★ **REMEMBER**
Passing score: 80% (16 of 20 correct). Any **safety** question answered wrong (Q3, Q6, Q14, Q19) is an automatic fail of the safety gate — re-teach and re-test before sign-off, regardless of total score. The Answer Key follows — no peeking until you're done!

Name: ______________________ **Date:** ______________ **Score:** ________ / 20 **Safety-gate Qs (all must be correct): 3, 6, 14, 19**

Q1. An acid copper bath is built from which three core ingredients?
A. Copper sulfate, sulfuric acid, and a trace of chloride B. Sodium cyanide and copper cyanide C. Boric acid and potassium chloride D. Nickel chloride and hydrochloric acid

Q2. (Short answer) What is the **water break test**, and what result tells you a part is clean?

Q3. **[SAFETY GATE]** If a **cyanide strike** is used, what must NEVER happen?
A. The part is entered live (current on) B. Acid contacts the cyanide bath, cyanide-wet parts, or cyanide rinses C. The strike is rinsed before acid copper D. The anodes are kept bagged

Q4. (Short answer) Why can't you plate acid copper *directly* onto steel? What do we do instead?

Q5. The chloride level in a **decorative** acid copper bath is held at about:
A. 5–8 g/L B. 500–800 ppm C. 50–80 ppm D. Zero — chloride is a contaminant

Q6. **[SAFETY GATE]** A part is **>40 HRC**. Per ASTM B850, the hydrogen-embrittlement relief bake is:
A. 200°F for 1 hour, any time within a week B. 500°F for 30 minutes C. No bake is needed for copper plating D. 375 ±25°F for at least 8 hours, started within 4 hours of the last plating step

Q7. (Short answer) Name the **three additives** in the acid copper system and, in a few words, what each does.

Q8. On the copper:acid "master lever," **low copper / high acid** gives you:
A. Superior throwing power — the PCB / high-throw regime B. Maximum decorative brightness and fastest plating C. Lower cathode efficiency D. No effect — the two act independently

Q9. (Short answer) What does the **anode phosphorus film** do, and what goes wrong if an anode is *filmless* versus *passivated*?

Q10. Acid copper's cathode efficiency is about:
A. 40–50% B. 60–70% C. ~95–100% D. Exactly 100%, always

Q11. (Short answer) Name the **two co-equal mainstream strike options** for steel, and one critical control you must apply when running the strike.

Q12. A Hull Cell panel that is **rough / nodular (sandpaper)** most likely means:
A. Particulate, or anode powder from high chloride / a filmless anode B. pH too high C. The bath is perfectly balanced D. Too much leveler and nothing else

Q13. A copper **strike** is best described as:
A. A thick, heavy copper build for corrosion protection B. An acid dip that removes oxide C. A passivate applied after plating D. A thin, tightly-bonded flash (~2.5–5 µm) for coverage and adhesion

Q14. (Safety) [SAFETY GATE] When diluting sulfuric acid, you must:
A. Add water to the concentrated acid quickly B. It doesn't matter which you add to which C. Always add acid to water, never water to acid D. Never dilute acid at all

Q15. (Short answer) Give the **two** reasons the post-plate rinse (Station 06) matters on a copper line.

Q16. The decorative "triad" (ASTM B456) is, in order:
A. Chromium → Nickel → Copper B. Copper → Chromium only C. Nickel → Copper → Zinc D. Copper (leveling) → Bright Nickel → Chromium

Q17. (Short answer) What does **chloride** do in the acid copper bath, and what happens if it goes too low versus too high?

Q18. Chloride in the bath is measured by:
A. A pH meter B. Titration with silver nitrate C. A Hull Cell only D. It cannot be measured

⚠ **SAFETY SECTION — MUST BE CORRECT TO CERTIFY**
Questions 3, 6, 14, and 19 cover hazards that can permanently harm you or others. A miss on any safety item means re-teach and re-test before sign-off.

Q19. (Short answer / Safety) [SAFETY GATE] List **four** pieces of PPE for the acid copper / activation station, name the procedure you must perform before any **rectifier** maintenance, and state the **cyanide rule**.

Q20. (Short answer) Acid copper levels and throws better than any common bath — but name **two** of its demanding traits (things it punishes if you get them wrong).

End of test. Hand to your trainer for grading.

FOR TRAINERS & SELF-CHECKERS

ANSWER KEY

SHORT-ANSWER RESPONSES DON'T NEED TO MATCH WORD-FOR-WORD — LOOK FOR THE KEY CONCEPTS IN BOLD

> ⚠ **HEADS-UP**
> Keep this page out of the trainee's copy. Questions **3, 6, 14, and 19** are safety-critical — any wrong answer there requires re-training on that topic before sign-off, regardless of total score.

Q1. A — Copper sulfate, sulfuric acid, and a trace of chloride (plus organic additives).

Q2. Dip and pull the part and watch the water. If it **sheets off in an unbroken, continuous film**, the part is clean. If it **beads up**, oil remains — reclean. (Free; use it every rack.)

Q3. B — Acid must never contact the cyanide bath, cyanide-wet parts, or cyanide rinses. Acid + cyanide releases **lethal HCN gas.**

Q4. Copper is more noble than steel, so it **immersion-deposits** (loose, non-adherent) on bare steel — causing **blistering.** Instead we apply a **copper strike** first (a thin adherent flash) so acid copper has a compatible surface. ($Cu^{2+} + Fe^0 \rightarrow Cu^0 + Fe^{2+}$.)

Q5. C — 50–80 ppm (decorative; 40–60 ppm for PCB/high-throw). Titrate with silver nitrate.

Q6. D — 375 ±25°F for at least 8 hours, started within 4 hours of the last plating step (ASTM B850 `[VERIFY rev]`).

Q7. Suppressor (carrier) — polarizes/refines grain, luster, macro-throw (needs chloride); **Accelerator (brightener, e.g. SPS)** — brightness, throw, via-fill; **Leveler** — smooths the surface, controls low-CD.

Q8. A — Superior throwing power (the PCB / high-throw regime). High copper / low acid is the decorative regime.

Q9. The **black Cu_3P film** keeps the anodes dissolving smoothly and suppresses copper powder. **Filmless** (new/unconditioned, or chloride too low) → sheds copper particles → **roughness. Passivated** (chloride/current too high, too little anode area) → stops dissolving → **copper starvation** and rising voltage.

Q10. C — ~95–100%.

Q11. The two co-equal mainstream strikes are the **cyanide copper strike** and the **alkaline non-cyanide copper strike.** Critical control (any one): **live/hot entry**, **complete coverage** before acid copper, or a proper **guard rinse** after. (Woods nickel and copper pyrophosphate are specialty strikes.)

Q12. A — Particulate, or anode powder from high chloride / a filmless anode.

Q13. D — A thin, tightly-bonded flash (~2.5–5 µm) whose job is coverage and adhesion, not thickness.

Q14. C — Always add acid to water, never water to acid ("do as you oughta — add acid to water").

Q15. (1) **Recover copper drag-out** (money) for return to the bath; (2) **keep copper and acid out of the downstream nickel / post-treatment** (copper poisons a nickel bath at a few ppm).

Q16. D — Copper (leveling) → Bright Nickel → Chromium.

Q17. Chloride partners with the **suppressor** to refine the deposit and maintains the **anode phosphorus film. Too low** (<~40 ppm): anodes polarize, deposit dull/coarse, poor throw. **Too high** (>~100 ppm): anodes passivate, shed copper powder, roughness.

Q18. B — Titration with silver nitrate.

Q19. PPE (any four): sealed chemical splash goggles (*not* safety glasses), face shield, chemical-resistant gloves, apron, chemical-resistant boots, respirator if ventilation inadequate. **Rectifier procedure: LOTO (Lockout/Tagout). Cyanide rule:** acid must NEVER contact cyanide (bath, parts, or rinses) — it releases lethal HCN gas.

Q20. Any two of: **won't plate active metals directly** (needs a strike); **extremely sensitive to particulate** (#1 roughness cause); **narrow chloride window** (50–80 ppm); **additives break down** and need carbon treatment; **needs phosphorized anodes with a maintained film**; **copper sulfate is toxic to aquatic life**; **acid mist needs ventilation**.

Answer distribution (MC): A=3 · B=2 · C=3 · D=3

> ★ **REMEMBER**
> A score of 16/20 passes — but every safety question (3, 6, 14, 19) must be correct to certify. If a trainee misses a safety item, re-teach it and re-test before sign-off. We don't gamble with people.

PRINT, FILL IN, SIGN

PLATING POSTERS

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CERTIFICATE OF COMPLETION

ACID COPPER PLATING TRAINING PROGRAM

This certifies that

____

OPERATOR NAME

has successfully completed the **Acid Copper Plating** training course — covering all eight process stages (Clean, Rinse, Activate, Strike, Plate, Rinse, Post-Treat/QC, and Final Rinse/Dry), bath chemistry and additive control, the copper strike and immersion-deposit rule, Hull Cell diagnostics, anode phosphorus-film management, hydrogen-embrittlement control, troubleshooting, and the safety practices for alkaline cleaners, mineral acids, the acid copper bath, and (where used) cyanide strike chemistry — and has passed the Completion Test with a score of __________ / 20, including all required safety competencies.

This operator is hereby cleared for **independent work** at the certified stations listed on the Operator Training Matrix.

Date of completion: ____________________ **Re-certification due:** ____________________

____

TRAINEE SIGNATURE

____

CERTIFYING TRAINER

____

SUPERVISOR

Training reference only. Verify all parameters against your chemistry supplier’s TDS, customer specifications, and applicable regulatory requirements before production use. Sulfuric acid causes severe burns; copper sulfate is toxic to aquatic life; cyanide strike chemistries are potentially lethal and must never contact acid. Always consult current SDS and comply with OSHA, EPA, REACH, and local regulations.

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